

## **Unit 2**

# **Financial Strategy**

# **Unit 2: Financial Strategy**

**Money and happiness**

**Money and human bias**

**Money happiness principles**

**Purpose, values and principles**

**Life stages and setting goals**

**Nominal and Real Rates of Return**

**Savings Calculations**

**Personal Finance Strategy**

# Discussion

**Sitting by yourself? Please move next to someone else!**

**Say “hello” and introduce yourself in groups of 2 or 3.**

**What things do you think will make a big difference to your feelings of ‘happiness’ and overall ‘life satisfaction’ over the next 40 years of your life?**

**How do you think your beliefs are different from those of your parents?**

# **General Advice Warning**

**The content of this Unit is not personal financial advice for you**

**It covers general principles**

**... taught as part of university course based on the Australian system.**

**It does not take into account your situation, objectives or needs.**

**You are responsible to consider whether it is suitable**

**... for your personal circumstances.**

**You should consider obtaining financial and tax advice yourself.**

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**Subtitles**

Closed Captions

**Playback Speed**

Slower

Faster

CC



# **Money and Happiness**

# **What do I mean by ‘money’ and ‘happiness’?**

**‘Money’ is deliberately ambiguous**

**... could be income, profit (saving), assets or equity (wealth)**

**Not really the emotional state of ‘pleasure’ (hedonia)**

**... more a sense of engagement, meaning or ‘life-satisfaction’ (eudaimonia)**

**Hedonia is like the hot flame of a campfire**

**... eudaimonia is like hot coals that are better for cooking!**

The terms ‘hedonia’ and ‘eudaimonia’ are from Aristotle (4th century BC) Nicomachean Ethics (pronounced nee-kuh-muh-kean).  
Eudaimonia is pronounced as you-die-monea

# Positive and negative affect

## Positive affect (PA)

**Tendency of someone to express positive emotions**

Cheerfulness, pride, interest, enthusiasm, energy, engagement, contentment and joy

## Negative affect (NA)

**Tendency of someone to express negative emotions**

Sadness, disgust, lethargy, fear, distress, anxiety and depression

## Both dispositional (trait-level) and situational (state-level)

Dispositional (trait-level) is your personality but also has a genetic component

Situational (state-level) depends on your situation and environment

Stefan Schmukle, Boris Egloff and Lawrence Burns (2002), 'The relationship between positive and negative affect in the Positive and Negative Affect Schedule', *Journal of Research in Personality* (based on Minnesota twins research)

David Lykken (1999). *Happiness: What studies on twins show us about nature, nurture, and the happiness set-point*

# **Happiness has four components**

- 1. Sustained positive emotion (positive affect)**
- 2. Recovery from negative emotion**
- 3. Pro-social behaviour and generosity**
- 4. Positive mind-wandering, mindfulness and ‘affective stickiness’**

# Causes of happiness

## 1. Inborn temperament

There is a genetic component to feeling life satisfaction and pleasure (positive affect)

## 2. Attitudes

Being optimistic and positive (positive affect), socially outgoing and generous to others

## 3. Activities

Actively engaged in meaningful work (or study), exercise and social engagement.

## 4. Demographics

Income, wealth, living situation, rule of law and political freedom

**... but the importance of each can vary greatly between individuals**

Madhu Mohanty (2014), 'What determines happiness? Income or attitude: Evidence from the U.S. longitudinal data', *Journal of Neuroscience*

Ed Diener (2008), 'Myths in the Science of Happiness and Directions for Future Research' in *The Science of Subjective Well-being*

David Lykken (1999). *Happiness: What studies on twins show us about nature, nurture, and the happiness set-point based on Minnesota twins research*  
by Auke Tellegen, David Lykken et al., (1988) 'Personality similarity in twins reared apart and together', *Journal of Personality and Social Psychology*

# Happiness and the brain

## Pleasure (hedonia) is easy to study

Pleasure or reward circuit of brain (nucleus accumbens)

Same for higher order pleasure (success, financial achievement)

... and lower order (food, sex)

Many recreational drugs damage the nucleus accumbens

## Life-meaning (eudaimonia) is more challenging to study

Default node network of brain (cingulate cortex, prefrontal cortex and temporal lobe)

These sections are active during 'wakeful rest' such as mindfulness exercises, prayer, meditation, daydreaming, introspection, self-appraisal and deeper thinking

Looking at your phone constantly (nomophobia) likely prevents 'wakeful rest'

Morten Kringelbach and Kent Berridge (2010), *The Neuroscience of Happiness and Pleasure*

Michael Bowen (2017) 'The science of happiness', *Sydney Morning Herald* (Video)

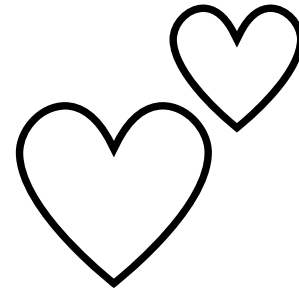
Jana Korunovska and Sarah Spiekermann-Hoff (2020), 'The Human Energy and Fatigue Constructs in Relation to Information and Communication Technologies: A Conceptual Framework', *Working Papers / Institute for IS & Society*

# Happiness and brain chemicals

## Dopamine

### The rewarding chemical

- Completing a task
- Doing self-care activities
- Eating food
- Celebrating little wins



## Oxytocin

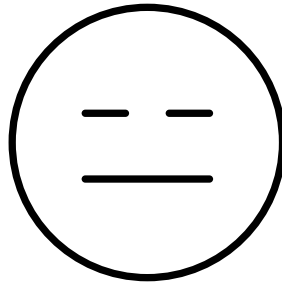
### The love hormone

- Playing with a pet
- Playing with a baby
- Holding hands
- Hugging your loved ones

## Serotonin

### The mood stabiliser

- Sun exposure
- Meditating
- Running
- Being in nature



## Endorphin

### The pain killer

- Laughing
- Exercising
- Dark chocolate
- Essential oils

# **Research into money and happiness is tricky**

- 1. Life satisfaction isn't the same thing as the 'emotion' of pleasure**
- 2. Life satisfaction is subjective and self-reported**
- 3. Wealthy people feel guilty reporting that they are not satisfied with life**
- 4. Detailed context information matters but studies use large samples**
- 5. Correlation does not mean causation**
- 6. Life satisfaction is multi-dimensional and complex**
- 7. Most research is based on income rather than wealth (equity)**

# Maslow's Hierarchy of Needs



Abraham Maslow (1943), 'A Theory of Human Motivation', *Psychological Review*.

Diagram from <https://www.simplypsychology.org/maslow.html>

# Money certainly helps with basic needs



Malahat Amani & Mohamad Javad Shabahang (2017) 'The relationship of self-efficacy and money attitudes with mental health: mediation through Maslow's hierarchy of needs', *International Journal of Culture and Mental Health*.

# Who should be happiest assuming ...

1. They are equally happy with all 'non-money' aspects of their lives
2. Money is good
3. More is better



**John**

Now

**\$20,000**



**Anna**

**\$100,000**



**David**

**\$500,000**

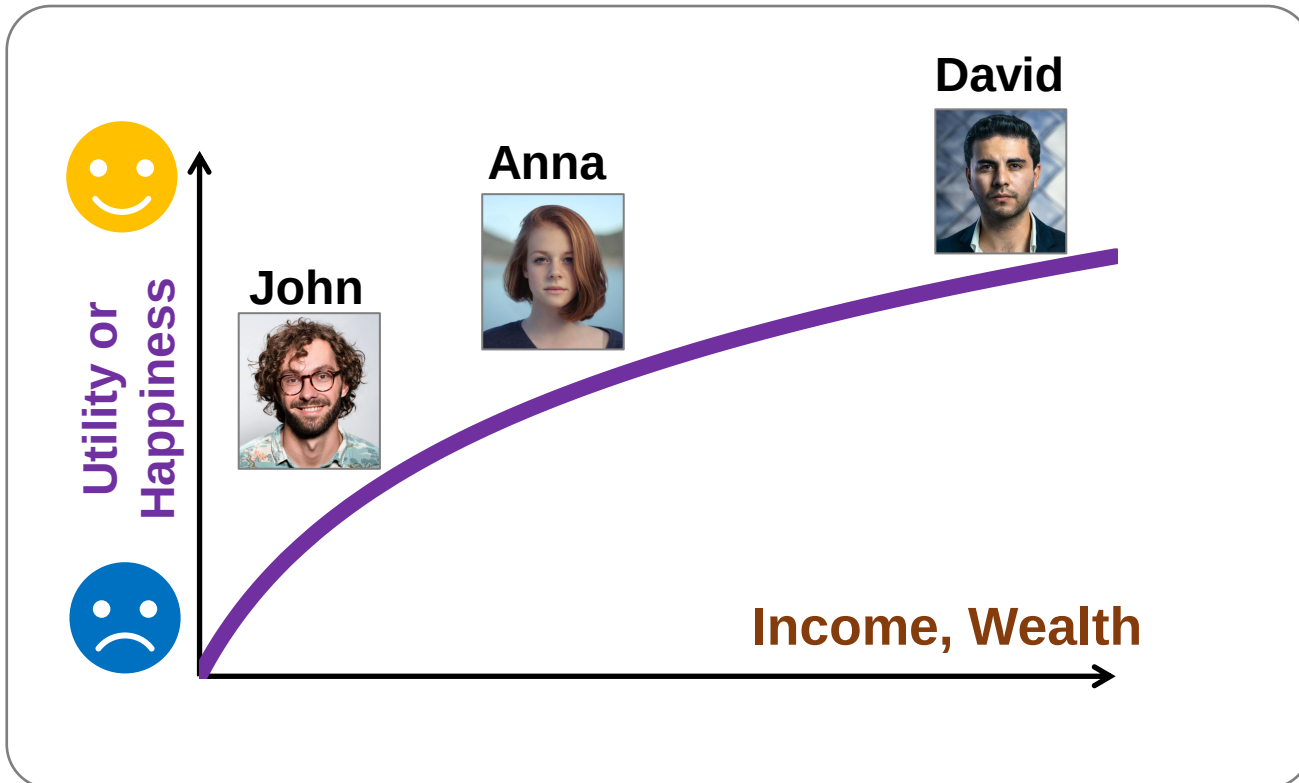


# Utility theory

More money allows you to satisfy more needs (utility)

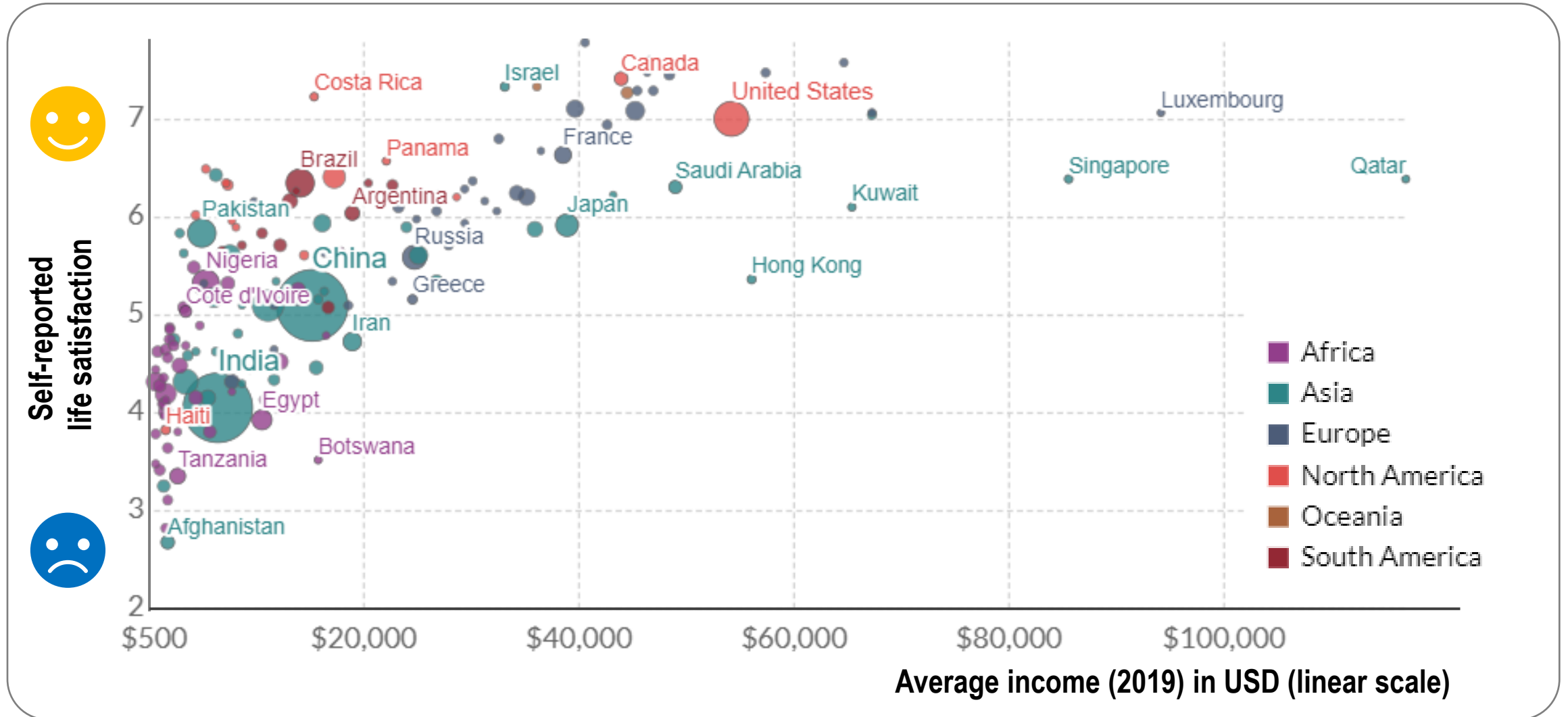
But each dollar gives less happiness than last one

This is called 'diminishing marginal utility'



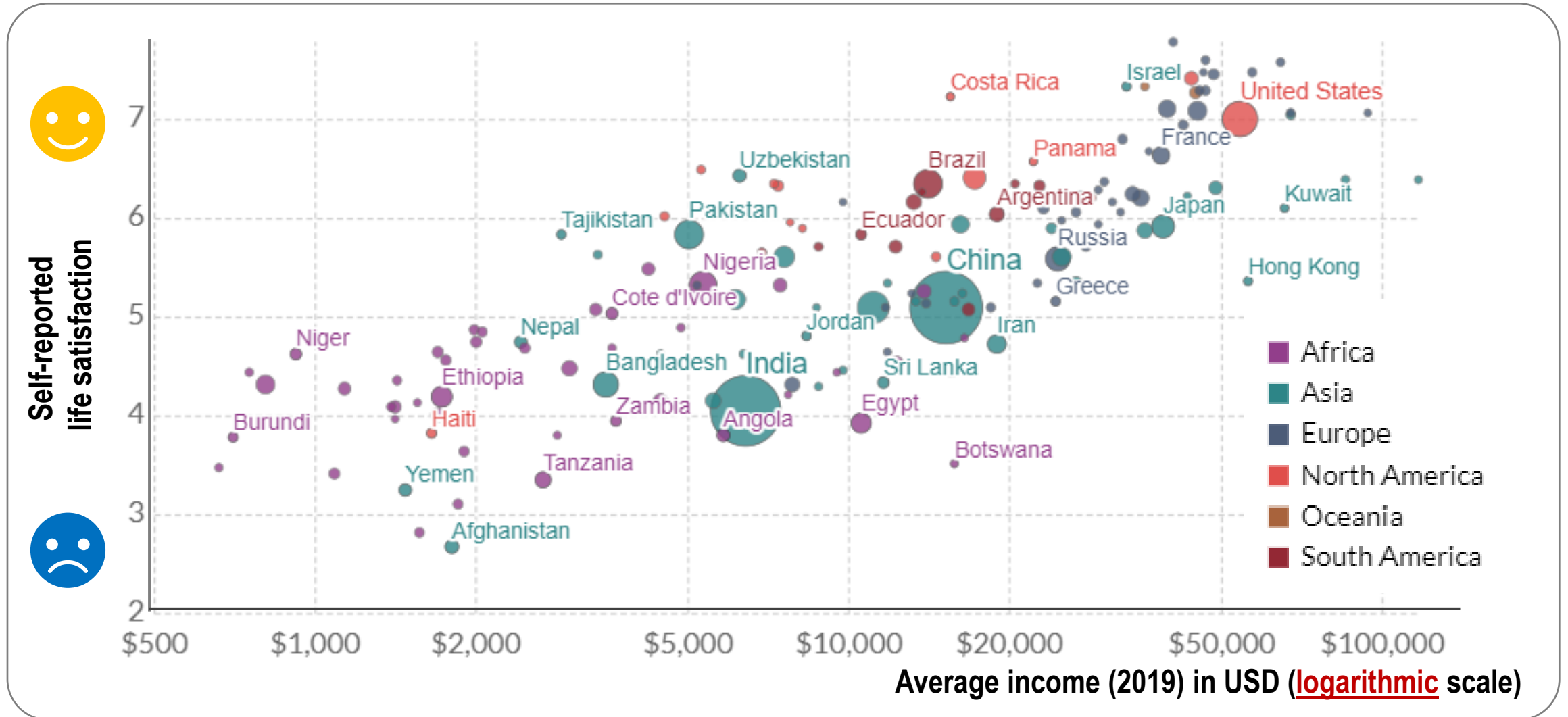
Utilitarianism and Utility Theory was developed by Jeremy Bentham (1789) and John Mill (1861) amongst others Ed Diener, Ed Sandvik, Larry Seidlitz and Marissa Diener (1993), 'The relationship between income and subjective well-being: Relative or absolute?' *Social Indicators Research*.

# Happiness and income



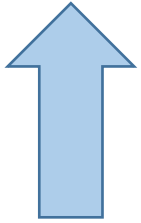
# Happiness and income

with logarithmic scale for average income



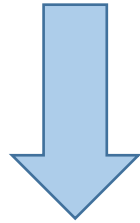
# Correlation does not prove causation

Happiness



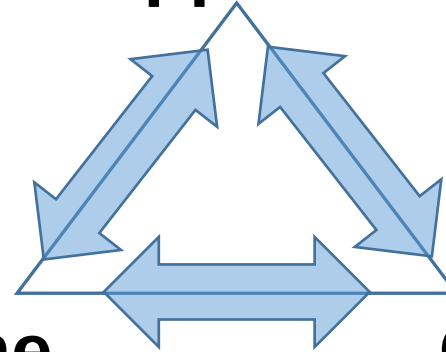
Income

Happiness



Income

Happiness



Income

Other

# Causes of average country happiness

Average income and perceived economic opportunities

The degree to which a society allows 'free choice'

Social trust and tolerance for diversity

Perceived 'fairness' in the distribution of income and wealth

Health, education and social security

Stable governance, rule of law and independent judiciary

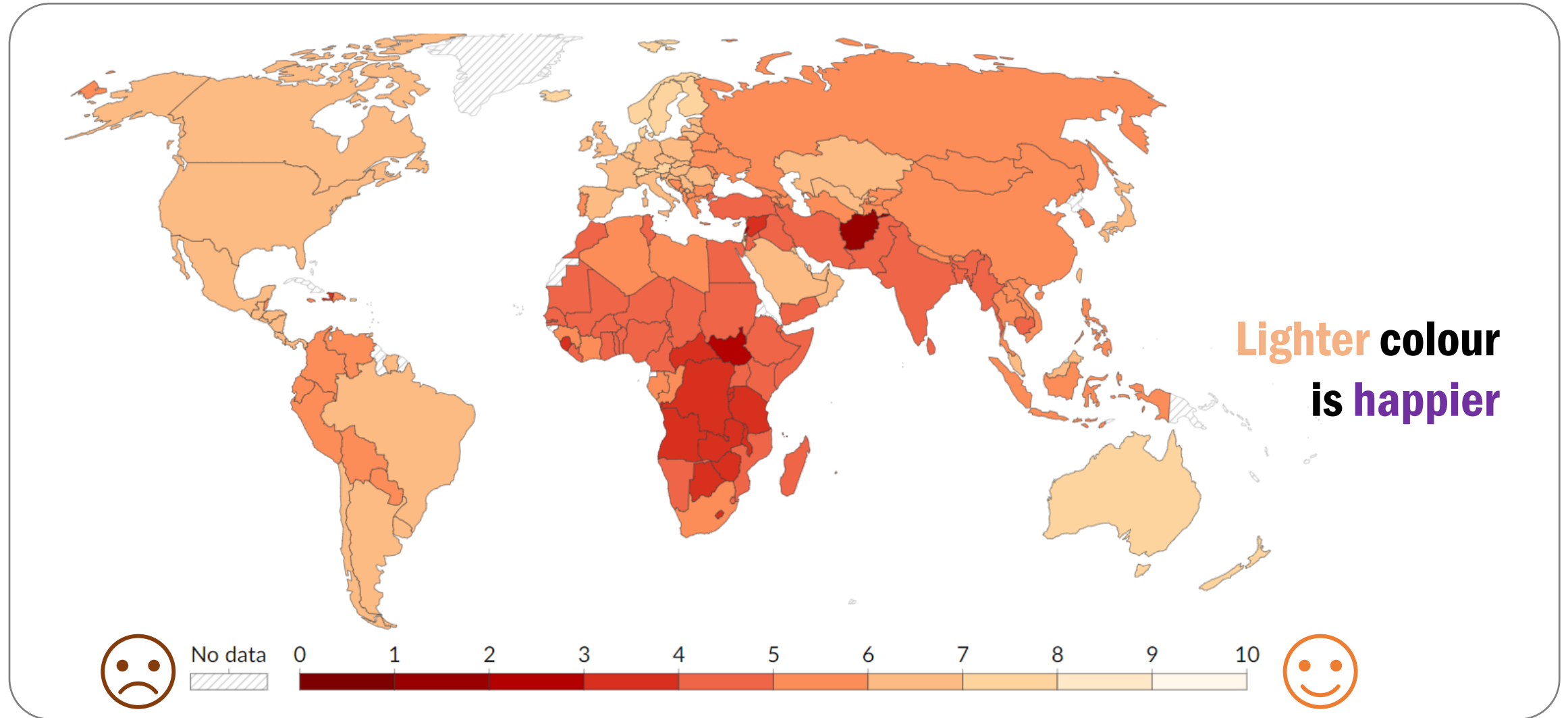
Political freedom and an independent media

Hans Rosling (2017) '[Hans Rosling: 200 years in 4 minutes – BBC News](#)' on Youtube (video)

Ronald Inglehart, Roberto Foa, Christopher Peterson and Christian Welzel (2008), '[Development, Freedom, and Rising Happiness: A Global Perspective \(1981-2007\)](#)', *Perspectives on Psychological Science*.

Ruut Veenhoven (2000), '[Freedom and Happiness: A Comparative Study in Forty-four Nations in the Early 1990s](#)' in *Culture and subjective well-being*.

# Happiness across the globe



# **Money and Human Bias**

# **A bat and ball**

**A ball costs 10 cents**

**A bat costs one dollar more than a ball**

**How much does it cost in total to buy a bat and a ball?**

# Humans are predictably irrational

## The brain uses heuristics when making decisions

It takes short-cuts due to limited time, information, ability, mental energy and attention

## These short-cuts often result in sub-optimal or irrational decisions

## Irrational decisions can be predictable in certain situations

A ball costs 10 cents

A bat costs one dollar more than a ball

How much does it cost in total to buy a bat and a ball?

Dan Ariely (2008), *Predictably Irrational: The Hidden Forces That Shape Our Decisions*

Dan Ariely (2008), 'Are we in control of our own decisions?', TED (Video)

Daniel Kahneman (2011), *Thinking Fast and Slow*

Daniel Kahneman (2016), 'Thinking, Fast and Slow. Daniel Kahneman (2011)', Youtube (Video)

# Who should be happiest assuming ...

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**John**

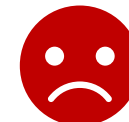


**Anna**



**David**

|            |          |           |             |
|------------|----------|-----------|-------------|
| Now        | \$20,000 | \$100,000 | \$500,000   |
| 1 year ago | \$10,000 | \$100,000 | \$1,000,000 |



# **The problem with classical utility theory**

**It assumes happiness is purely rational based on objective states**

**It assumes we don't compare with social peers (relative income)**

**It assumes we don't compare with the past (gains or losses)**

**It assumes we don't compare actual versus expected (gains and losses)**

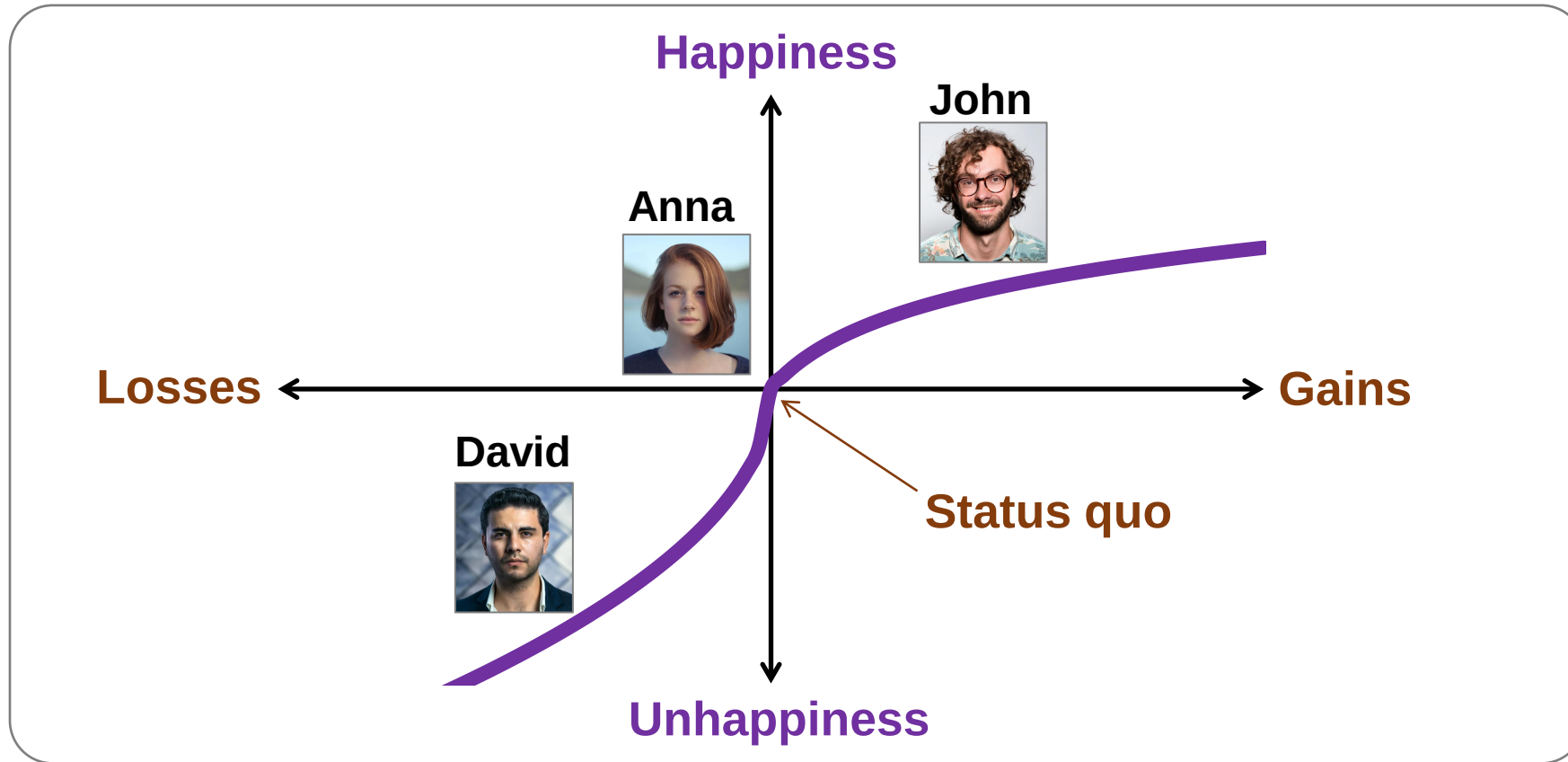
**We react more to perceived gains and losses against a 'status quo'**

**We over-react to perceived losses**

**We especially over-react to the prospect of a perceived future loss**

# Prospect Theory

Gains and losses against a perceived 'status-quo' are what matter



Daniel Kahneman and Amos Tversky (1979), 'Prospect theory: An analysis of decision under risk', *Econometrica*  
Daniel Kahneman was awarded the Nobel Prize in Economics in 2002 (Amos died in 1996 unfortunately)

**Which creates more happiness? Why?**

**Win \$100**

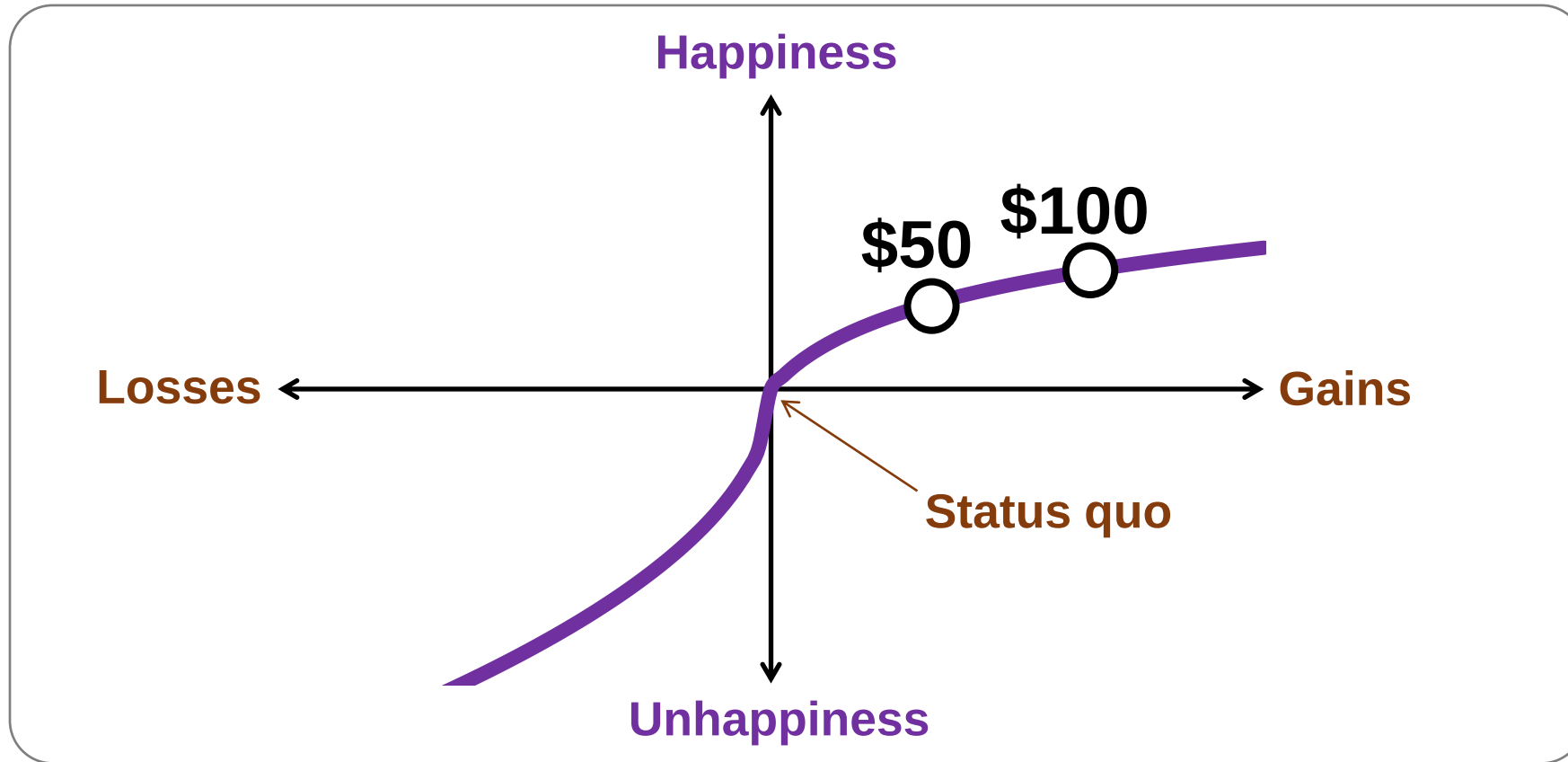
**Win \$50**

... followed a week later by

**Win \$50**

# Prospect Theory with separate gains

Gains and losses against a perceived 'status-quo' are what matter



Daniel Kahneman and Amos Tversky (1979), 'Prospect theory: An analysis of decision under risk', *Econometrica*  
Daniel Kahneman was awarded the Nobel Prize in Economics in 2002 (Amos died in 1996 unfortunately)

**Which creates more unhappiness? Why?**

**\$100 Fine**

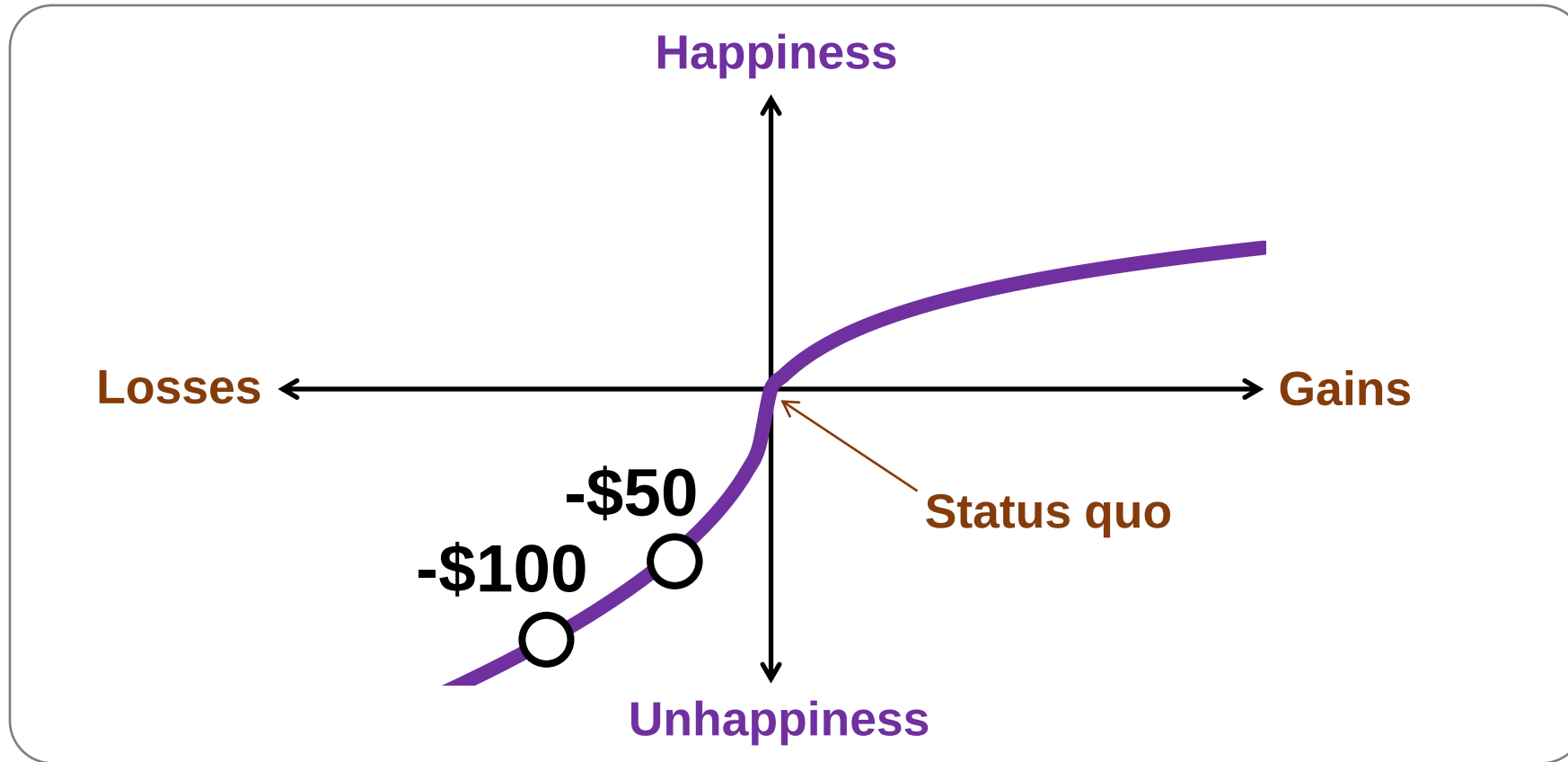
**\$50 Fine**

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# Prospect Theory with separate losses

Gains and losses against a perceived 'status-quo' are what matter



Daniel Kahneman and Amos Tversky (1979), 'Prospect theory: An analysis of decision under risk', *Econometrica*  
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# **Prospect Theory and happiness**

- 1. More is not necessarily better**
- 2. What matters is gains and losses against status quo**
- 3. We choose status quo and this 'frames' comparisons**
- 4. Split up good experiences will normally yield more happiness**
- 5. Combine bad experiences will normally minimise unhappiness**
- 6. Avoid unfavourable comparisons**
- 7. Be aware of irrationality with the prospect of future losses**

# **Adaptation spoils happiness**

**Our adaptability is the key to the success of our species**

**It also makes happiness illusive**

**We adapt to good things in our lives**

Our home, car, food, people, devices, experiences ...

**They become part of the status quo**

**And then no longer trigger gains and losses**

**This is called the ‘Hedonic Treadmill’**

# **We are hamsters on a hedonic treadmill**

**I will be happy when ...**

# **Quote from Stacie Orrico**

**I've got it all, but I feel so deprived  
I go up, I come down and I'm emptier inside  
Tell me what is this thing that I feel like I'm missing  
And why can't I let it go.**

**There's gotta be more to life  
Than chasing down every temporary high to satisfy me  
Cause the more that I'm...  
Trippin' out thinkin' there must be more to life  
Well it's life, but I'm sure... There's gotta be more.**

# **Quote from King Solomon** 10<sup>th</sup> Century B.C.

**“Meaningless! Meaningless!”, says the Teacher.**

**“Utterly meaningless! Everything is meaningless.”**

**What do people gain from all their labors at which they toil under the sun? ...**

**... The eye never has enough of seeing, nor the ear its fill of hearing.**

**What has been will be again, what has been done will be done again;  
there is nothing new under the sun ...**

**... I denied myself nothing my eyes desired; I refused my heart no pleasure.  
My heart took delight in all my labor, and this was the reward for all my toil.  
Yet when I surveyed all that my hands had done and what I had toiled to  
achieve, everything was meaningless, a chasing after the wind;  
nothing was gained under the sun.**

# What influences the rate of adaptation?

## 1. Novelty

... we have not experienced the event before

## 2. Surprise

... we didn't expect the event to happen the way it did

## 3. Uncertainty

... we are not sure how to interpret the event

## 4. Variability

... the experience keeps changing over time

**They force us to pay attention and evaluate gains or losses**

# Human bias and money happiness

**Are we acting irrationally due to losses incurred or feared?** Loss Aversion

**Do we fail to take action and make changes?** Status-Quo Bias

**Do we overly attribute past successes to skill?** Overconfidence Effect

**Have we 'fallen in love' with one option?** Affect Bias

**Have we been influenced by stories or analogies?** Saliency Bias

**Have credible alternatives really been considered?** Confirmation Bias

... continued next slide

# Human bias and money happiness

... continued from previous slide

**Have we created arbitrary 'buckets' for expenses?** Mental Accounting Bias

**Would a different decision be made in a year?** Availability Bias

**Where do the numbers come from?** Anchoring Bias

**Are we influenced by the attractiveness of the messenger?** Halo Effect

**Are people biased by past decisions?** Sunk Cost Bias and Endowment Effect

# Money personality profiles

## 1. **Achieving money worshipers**

Worship money as their success and budget money carefully

## 2. **Careless money admirers**

Value financial success but do not budget money carefully

## 3. **Apathetic money managers**

Money is not evil nor a motivator and have high intrinsic job and life satisfaction

## 4. **Money repellent individuals**

Consider money to be evil and is not a measure of success

Based on Love of Money Scale (LOMS) and Money Ethic Scale (MES)

Thomas Tang, David Tang and Roberto Luna-Arocas (2005), 'Money profiles: the love of money, attitudes, and needs', *Personnel Review*

# **Summary**

**There are many problems with classical thinking (rational)**

**Humans are predicably irrational (heuristics)**

**Behavioural economics and finance takes this into account**

**A key behavioural model is 'Prospect Theory'**

**We assess gains or losses against a perceived status quo**

**Adaptation spoils happiness (hedonic treadmill)**

**It is helpful to be able to 'name' different forms of human bias**

**It can be helpful to link someone to a 'money profile'**

# **Money Happiness Principles**

# **1. Define your own values and goals**

**Focus on achieving intrinsic goals consistent with your own values**

**... rather than extrinsic approval from others or financial success**

**... especially for ‘achieving money worshipers’**

**Tracking and measuring success adds to feeling of ‘achievement’**

Robert Emmons (1999). The psychology of ultimate concerns: Motivation and spirituality in personality.

Tim Kasser and Richard Ryan (2001), ‘Be Careful What You Wish For: Optimal Functioning and the Relative Attainment of Intrinsic and Extrinsic Goals’, in Life Goals and Well-Being: Towards a Positive Psychology of Human Striving.

## **2. Avoid placing too much emphasis on ‘things’**

**Placing too high importance on accumulating income and wealth  
as a source of happiness (materialism) can be detrimental  
... we adapt and can end up on a hedonic treadmill**

Van Boven, L. (2005), 'Experientialism, materialism, and the pursuit of happiness', Review of General Psychology.

Joseph Sirgy (1997), 'Materialism and Quality of Life', Social Indicators Research.

Daniel Kahneman, Alan Krueger, David Schkade, Norbert Schwarz and Arthur Stone (2004), 'A Survey Method for Characterizing Daily Life Experience: The Day Reconstruction Method', Science.

Carol Nickerson, Norbert Schwarz, Ed Diener and Daniel Kahneman (2003), 'Zeroing in on the Dark Side of the American Dream: A Closer Look at the Negative Consequences of the Goal for Financial Success', Psychological Science

# 3. Invest in relationships

**Most happiness is determined by quality and breadth of relationships**

Romantic partner, children, family, friends, work ...

**Each interaction is unique ... making it difficult to adapt**

**... forcing higher cognition ... triggering gains and losses**

**They are also an important component of pro-social behaviour**

Cassie Mogilner and Michael Norton (2016), 'Time, money, and happiness', *Current Opinion in Psychology*

Ed Diener and Martin Seligman (2002), 'Very Happy People', *Psychological Science*.

Richard Easterlin (2001), 'Income and Happiness: Towards a Unified Theory', *The Economic Journal*.

Carol Nickerson, Norbert Schwarz, Ed Diener and Daniel Kahneman (2003), 'Zeroing in on the Dark Side of the American Dream: A Closer Look at the Negative Consequences of the Goal for Financial Success', *Psychological Science*

# **4. Invest in experiences**

**We adapt quickly to intransient physical things**

**Experiences require higher cognition**

**... and are more likely to trigger gains and losses**

**Young people gain more happiness from extraordinary experiences**

**Older people gain more happiness from ordinary experiences**

**Experiences with other people have the most significant impact**

Cassie Mogilner and Michael Norton (2016), 'Time, money, and happiness', *Current Opinion in Psychology*

Amit Bhattacharjee and Cassie Mogilner (2014), 'Happiness from Ordinary and Extraordinary Experiences', *Journal of Consumer Research*.

# 5. Help others rather than yourself

Being generous with time and gifts with others  
provides more happiness than most people realise!

Michael Norton (2011) 'How to buy happiness', *TED (Video)*

Elizabeth Dunn, Lara Aknin and Michael Norton (2008) 'Spending Money on Others Promotes Happiness', *Science*.

Elizabeth Dunn and Michael Norton (2014), *Happy Money*.

# **6. Focus more on time than money**

**Focus more on spending time on happiness**

**... rather than spending money on happiness**

**Start by making a list of the things that bring you the most joy in life!**

# **7. Maintain some financial slack**

**Financial slack (emergency savings) provides ‘peace of mind’  
when experiencing an unexpected shock or crisis**

**Keep an emergency fund with at least 6+ months of living expenses!**

# **8. Actively work to serve others**

**Active employment is a significant contributor to happiness**

... and it doesn't necessarily need to be 'paid' employment

**This builds relationships with others**

**... creates a feeling of outward-focussed purpose.**

# **9. Buy regular small pleasures over big ones**

**Prospect theory predicts that separating gains  
will elicit a greater perceived gain**

**Although for young people, this needs to be balanced with  
extraordinary experiences being preferred over ordinary ones**

# **10. Enjoy 'free' anticipation** delay gratification

**The person who buys a good gets X units of pleasure**

**The person who anticipates then buys it gets  $X + Y$  units of pleasure**

**Delayed gratification is highly correlated with health, wealth & happiness**

Cassie Mogilner and Michael Norton (2016), 'Time, money, and happiness', Current Opinion in Psychology

Ying-Yao Cheng, Paichi Pat Shein and Wen-Bin Chiou (2011), 'Escaping the impulse to immediate gratification: The prospect concept promotes a future-oriented mindset, prompting an inclination towards delayed gratification', British Journal of Psychology.

# 11. Happiness is in the details

**Day to day life experience  
have biggest impact  
on happiness**

**Avoid seeking happiness  
in abstract long-term plans  
as they tend to be  
light on detail**

The School of Life (2016), 'Why Small Pleasures are a Big Deal', Youtube (Video)

Daniel Kahneman, Alan Krueger, David Schkade, Nobert Schwarz and Arthur Stone (2004), 'A Survey Method for Characterizing Daily Life Experience: The Day Reconstruction Method', Science.

# 12. Practice contentment and thankfulness

**‘Counting blessings’ slows down  
the rate of adaptation  
while also eliciting a perceived gain.**

**‘Minimal social interactions’ of greeting, thanking and conversing.  
Consider starting a ‘gratitude journal’**

Esra Ascigil, Gul Gunaydin and Erdal Aydin (2023), ‘Minimal Social Interactions and Life Satisfaction: The Role of Greeting, Thanking and Conversing’, Social Psychological and Personality Science

Mariano Rojas (2011), ‘Contentment and Affect in the Estimation of Happiness’, Social Indicators Research.

Tim Kasser (2011), ‘Can Thrift Bring Well-being? A Review of the Research and a Tentative Theory’, Social and Personality Psychology Compass

Courtney Ackerman (2017), ‘Gratitude Journal: 66 Templates and Ideas for Daily Journaling’, PositivePsychology.com

# 13. Avoid unfavourable comparisons

**Prospect theory predicts that choosing an unfavourable ‘status quo’ will trigger off an unnecessary and adverse sense of loss.**

Ed Diener and Richard Lucas (2000), ‘Explaining differences in societal levels of happiness: Relative standards, need fulfillment, culture, and evaluation theory’, *Journal of Happiness Studies*

# 14. Look after your health

**There is a clear connection between your mind and body.**

**Sleep** (quantity and quality)

**Nutrition** (food and drink)

**Physical activity** (movement and sunshine)

# **15. If you pursue happiness you won't find it**

**Hedonism is a hedonic treadmill**

**We are most happy when  
we are 'lost' in a task and  
lose sense of time**

**We consistently overestimate  
the happiness derived from leisure**

# PERMA+ model of well-being

**P**ositive Emotions Hope, interest, joy, love, compassion, pride, amusement, gratitude

**E**ngagement Loss of self-consciousness and complete absorption in activity (state of 'flow').

**R**elationships Feeling supported, loved and valued by family, friends and the community.

**M**eaning Belonging and serving something greater than ourselves. Feeling a sense of 'purpose'.

**A**chievement Accomplishment from mastering something and working towards goals.  
or accomplishment

**+** Nutrition, Physical Activity and Sleep The connection between mind and body.

Martin Seligman (2012) *Flourish: A visionary new understanding of happiness and well-being*

Martin Seligman (2018) 'PERMA and the building blocks of well-being' *The Journal of Positive Psychology*

Melissa Madeson (2017) 'Seligman's PERMA+ Model Explained: A Theory of Wellbeing', [PositivePsychology.com](https://www.positivepsychology.com)

# **Rita Mae Brown**

Hiss of Death (2011)

**“Happiness is pretty simple:  
someone to love,  
something to do,  
something to look forward to.”**

# Break

# **Purpose, Values and Principles**

# **A bit macabre ... but very important!**

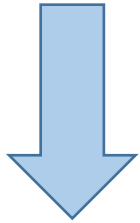
**You have just passed away at age 85**

**A close life-long friend is giving a eulogy at your funeral**

**List 5 to 10 points that they say about your character and life?**

# What are values, principles and purpose?

## Values

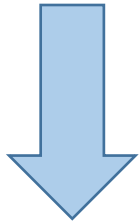


**A quality or standard of behaviour that tell us what is ‘good’**

Expressed in general terms and focusses on what we value

Often expressed in one or two words, such as “thriftiness” or “honesty”

## Principles



**A specific rule governing behaviour that tell us what is ‘right’**

Specific and applied statements outlining how we may (or may not) achieve our values

Often expressed in statements as “I’ll never ...” or “I’ll always ...”

## Purpose

**Provides our reason for being**

It explains why our values and principles are important

The Ethics Centre (2020), ‘Purpose, Values, Principles – An Ethics Framework’ on Youtube (Video)

The Ethics Centre (2019), ‘Why purpose, values, principles matter’ on ethics.org.au

Simon Longstaff, Katherine Hunt and Carolyn Tate (2020), Everyday Ethics for Financial Advisers

# Examples of values and principles

| Value                       | Principle                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Thriftiness</b>          | I'll always practice contentment and avoid buying things just because they are 'new' or to impress others.                                                                                                                 |
| <b>Open-mindedness</b>      | I'll practice listening, compassion and empathy so that I can understand other people's thinking and behaviour.                                                                                                            |
| <b>Fairness</b>             | I'll never treat people unfairly by making judgements based on their gender, sexual orientation, age, race, cultural background or appearance.                                                                             |
| <b>Honesty</b>              | I'll never lie, even if it is to prevent harm, hurt feelings or to avoid an uncomfortable situation. I'll always be honest in a way that is also loving, compassionate and has the best interests of other people in mind. |
| <b>Relationships matter</b> | I'll never be unfaithful to my friends or partner. I will always put the long-term interests of people ahead of short-term financial gain.                                                                                 |

# Why do values matter?

1. Conducive for happiness via intrinsic motivation
2. Important for determining ethical frameworks
3. Helpful for making ethical decisions involving trade-offs
4. Influence pro-social behaviour and generosity
5. Help you to set your long-term life and financial goals
6. Assist you to choose a career and develop a career strategy
7. Influence strategies, tactics and product selection to achieve goals

Simon Longstaff, Katherine Hunt and Carolyn Tate (2020), *Everyday Ethics for Financial Advisers*

The Ethics Centre (2019), 'Why purpose, values, principles matter' on [ethics.org.au](https://ethics.org.au)

Robert Emmons (1999). *The psychology of ultimate concerns: Motivation and spirituality in personality*.

Tim Kasser and Richard Ryan (2001), 'Be Careful What You Wish For: Optimal Functioning and the Relative Attainment of Intrinsic and Extrinsic Goals', in *Life Goals and Well-Being: Towards a Positive Psychology of Human Striving*.

# **How many values should I have?**

**There is a 'goldilocks principle' with the number of values**

**Too few values ... is overly simplistic for the complexity of life decisions**

**Too many values ... and they are unlikely to be meaningful and will conflict**

**For your financial plan ... choose between 3 and 10**

Recommended between 5 and 7 for most people

# Identifying your core values

1. Existential or spiritual beliefs
  2. Cultural and family background
  3. Think of 3 to 6 people that you admire or love. Why?
  4. Look back ... what were the most bitter or sweet moments? Why?
  5. Look forward ... what will people say at your funeral? Why?
  6. Start narrow: Short Schwartz Value Survey (10 values)
  7. Go wide: Values Checklist by Russ Harris (50+ values)
  8. Free online inventories: Life Values Inventory by Crace and Brown
  9. Observe yourself, talk to trusted friends and family and learn
- } provided under 'Course Resources'

Life Values Inventory available at <https://www.lifevaluesinventory.org/>

Shalom Schwartz (2012), 'An Overview of the Schwartz Theory of Basic Values', *Online Readings in Psychology and Culture*

Meg Selig (2018), '6 Ways to Discover and Choose your Core Values', *Psychology Today*

# Schwartz theory of basic values

## 1. Universalism

Protect environment, unity with nature, social justice, wisdom, equality, inner harmony

## 2. Tradition

Devout, accept portion in life, humble, moderate, respect for tradition, detachment

## 3. Conformity

Politeness, honouring parents and elders, obedient, self-disciplined

## 4. Achievement

Successful, capable, ambitious, influential, intelligent, self-respectful

## 5. Stimulation

Daring, an exciting life, a varied life

Shalom Schwartz (1992), 'Universals in the Content and Structure of Values: Theory and Empirical Tests in 20 Countries' in *Advances in Experimental Psychology* and Jasper Kenter et al., (2015), 'What are shared and social values of ecosystems', *Ecological Economics*.

# Schwartz theory of basic values

## 6. Benevolence

Helpful, honest, forgiving, loyal, responsible, friendship, spiritual, mature love, meaningful life

## 7. Security

Clean, national security, social order, family security, reciprocation of favours, healthy, belonging

## 8. Self-direction

Creativity, curious, freedom, choosing own goals, independent

## 9. Power

Social power, authority, wealth, preserving public image, social recognition

## 10. Hedonism

Pleasure, enjoying life

Shalom Schwartz (1992), 'Universals in the Content and Structure of Values: Theory and Empirical Tests in 20 Countries' in *Advances in Experimental Psychology* and Jasper Kenter et al., (2015), 'What are shared and social values of ecosystems', *Ecological Economics*.

# Russ Harris' Value Checklist

## 1. **Acceptance**

to be open to and accepting of myself, others, life etc

## 2. **Adventure**

to be adventurous; to actively seek, create, or explore novel or stimulating experiences

## 3. **Assertiveness**

to respectfully stand up for my rights and request what I want

## 4. **Authenticity**

to be authentic, genuine, real; to be true to myself

## 5. **Beauty**

to appreciate, create, nurture or cultivate beauty in myself, others, the environment etc

**... continued for another 50+**

# **Values can change over time**

**Changes in values tend not to be ‘chaotic’**

**... but the composition of your top 5 – 7 and their importance do change**

**Life-changing events (rather than age) causes these changes**

**Increase in importance of any one value is associated with**

**... an increase in importance of compatible values**

**... a decrease in importance of conflicting values**

# Life principles

**Life principles explain ‘how’ you plan to live-out your values**

**Statements often expressed as “I’ll always ...” or “I’ll never ...”**

**They are applications of the value ... for example**

## **Honesty**

I’ll never lie, even if it is to prevent harm, hurt feelings or to avoid an uncomfortable situation.

I’ll always be honest in a way that is also loving, compassionate and has the best interests of other people in mind.

The Ethics Centre (2020), ‘Purpose, Values, Principles – An Ethics Framework’ on Youtube (Video)

The Ethics Centre (2019), ‘Why purpose, values, principles matter’ on [ethics.org.au](https://ethics.org.au)

Simon Longstaff, Katherine Hunt and Carolyn Tate (2020), *Everyday Ethics for Financial Advisers*

# Life purpose

**An overall statement of the ‘why’ behind your values and principles**

**You can have different ‘purpose’ statements for work and personal life**

**‘Work related’ life purpose statements can be included on a resume**

The Ethics Centre (2020), ‘Purpose, Values, Principles – An Ethics Framework’ on Youtube (Video)

The Ethics Centre (2019), ‘Why purpose, values, principles matter’ on [ethics.org.au](https://ethics.org.au)

Simon Longstaff, Katherine Hunt and Carolyn Tate (2020), *Everyday Ethics for Financial Advisers*

# **Summary**

**Values tell is 'what' is good**

**Principles tell us 'how' we may achieve our values**

**Purpose explains 'why' our values and principles are important**

# Life Stages

# **Discussion**

**Is it worth having a long-term plan for your life  
when there is so much uncertainty about the future?**

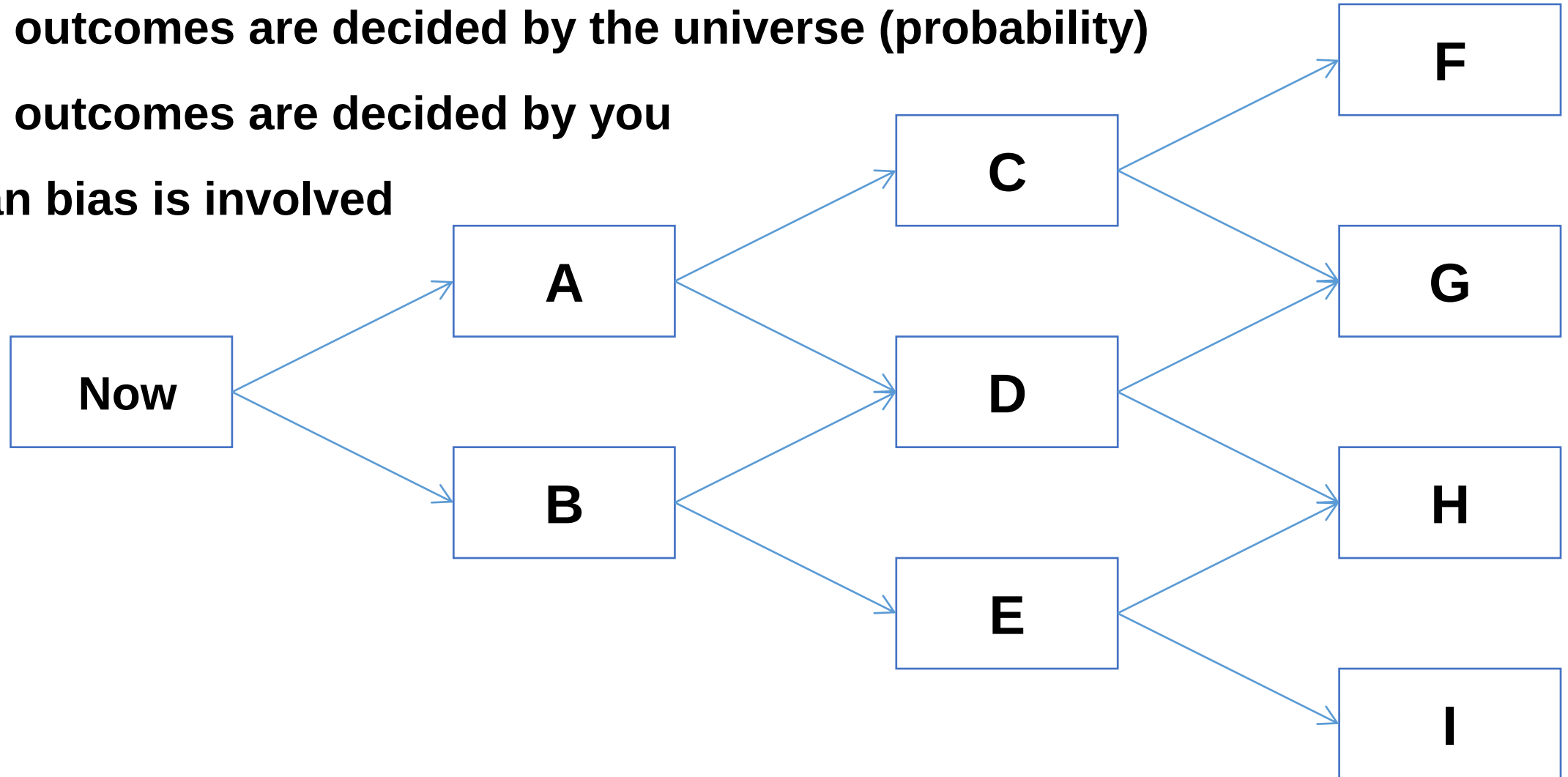
**What do you think are some of the benefits of long-term planning?  
What are some of the weaknesses?**

# Your life is a decision tree

Some outcomes are decided by the universe (probability)

Some outcomes are decided by you

Human bias is involved



# **The definition of ‘wisdom’** according to Andrew Hingston

**To understand with clarity (free from human bias)**

- 1. our current situation**
- 2. the best ‘reasonable’ future outcome given our current situation**
- 3. the actions we can take to achieve that outcome**
- 4. the probabilities of outcomes given our actions**

We are all **biased** in our beliefs about our situation, outcomes and probabilities

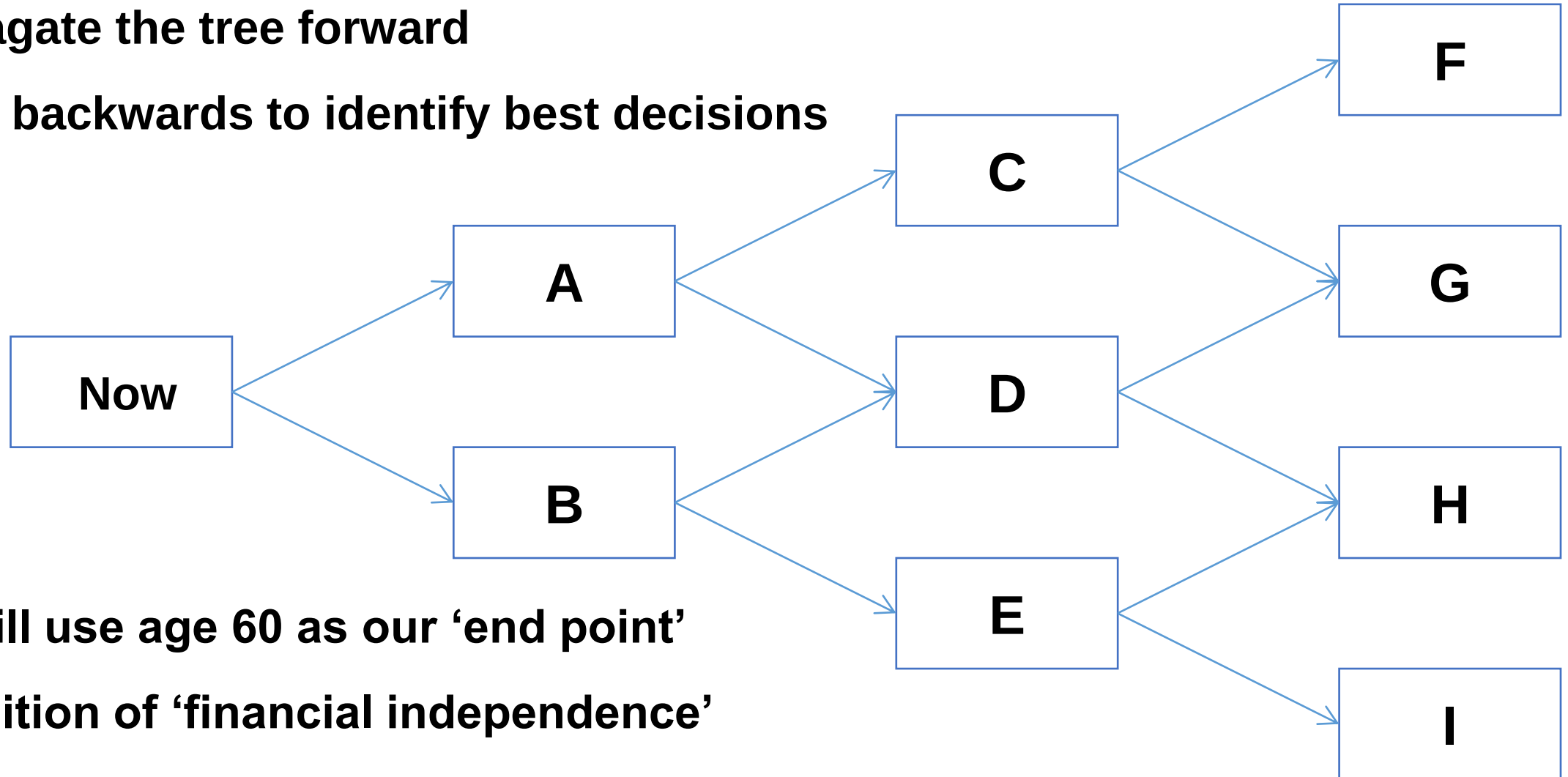
We often fail to identify actions that we can take

... or lack the self-control to implement them

# Think forward then solve backward

Propagate the tree forward

Solve backwards to identify best decisions



We will use age 60 as our 'end point'

A position of 'financial independence'

# **What do I mean by financial independence?**

**A situation in which your investments generate enough cash flow to cover your living expenses indefinitely**

**You no longer need to do paid work**

**... but can choose to do paid or unpaid work**

**This is the main focus of the next Unit of the course**

# **Important activity**

**Think back to your values and the principles of money and happiness**

**Now imagine that you are age 60 and financially independent. Describe ...**

- 1. Your closest relationship (romantic or otherwise)**
- 2. Your relationships with your immediate family (children, parents, siblings)**
- 3. Any paid or unpaid work to keep you engaged and serving others**
- 4. Any hobbies or interests including with whom you practise them**
- 5. Where you are living and with whom you are living**
- 6. A typical holiday and with whom you are on holiday**

# Important activity

Think back to your values and the principles of money and happiness

Now imagine that you are age 60 and financially independent. Describe ...

1. Your closest relationship (romantic or otherwise)
2. Your relationships with your immediate family (children, parents, siblings)
3. Any paid or unpaid work to keep you engaged and serving others
4. Any hobbies or interests including with whom you practice them
5. Where you are living and with whom you are living
6. A typical holiday and with whom you are on holiday

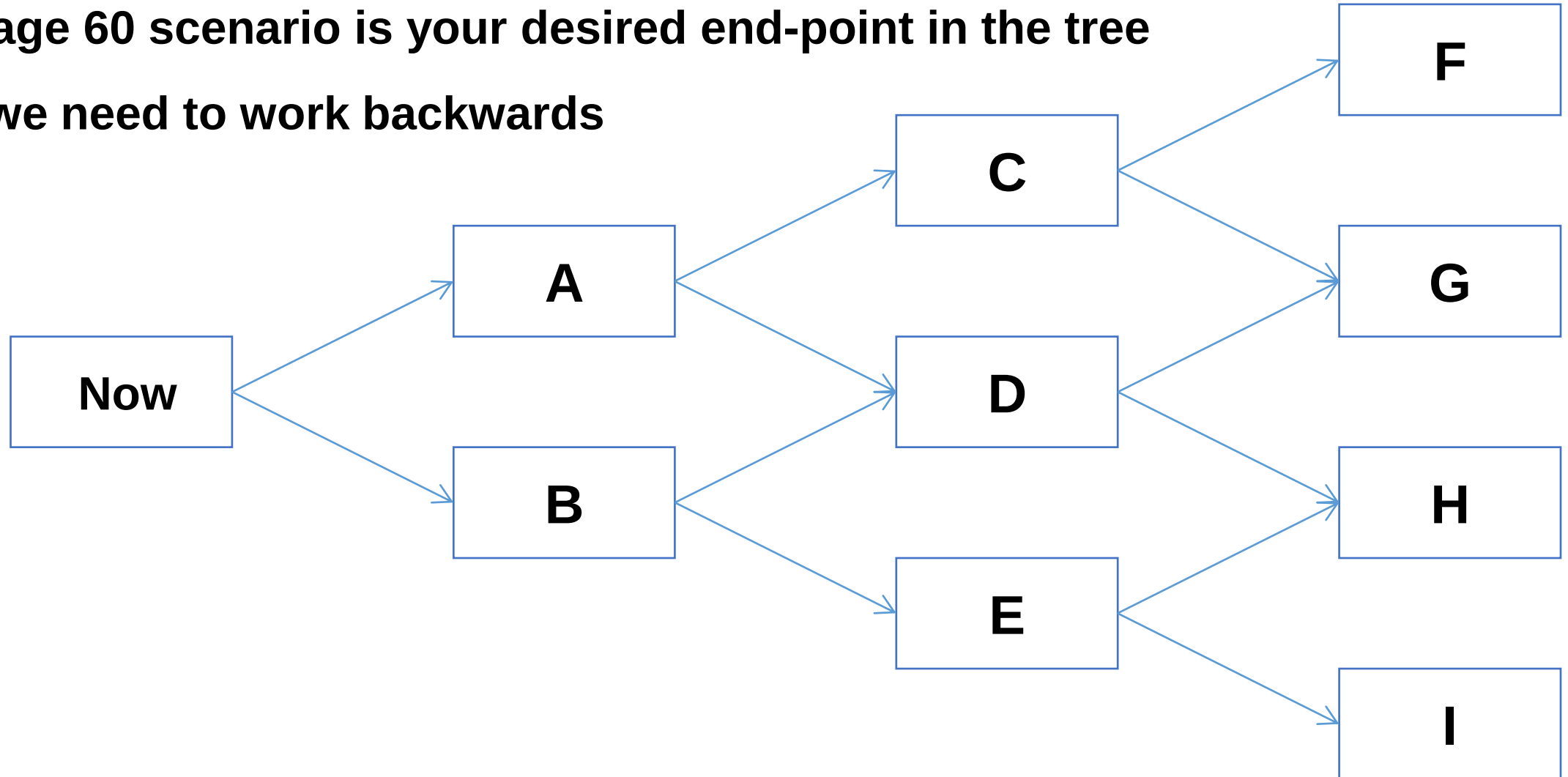
Assuming you own your own dwelling and ignoring inflation ...

... what is the minimum income would you need to fund that lifestyle?

# Now work backwards along the tree ...

Your age 60 scenario is your desired end-point in the tree

Now we need to work backwards



# There are different paths through the tree ...

| LIFE PATH 1                              | LIFE PATH 2                               |
|------------------------------------------|-------------------------------------------|
| Student (19 – 23)                        | Student (19 – 23)                         |
| Single worker (24 – 30)                  | Single worker (24 – 30)                   |
| Couple with dependent children (31 – 50) | Start first business (31 – 50)            |
| Couple with adult children (51 – 60)     | Start second business (51 – 60)           |
| Financial independent couple (61 – 85)   | Financial independent investors (61 – 85) |

# Q: Your 'draft' life path

- a) Define between 3 and 7 'life stages'
- b) Add some rough age ranges for each

| <b>LIFE PATH 1</b>                              |
|-------------------------------------------------|
| <b>Student (19 – 23)</b>                        |
| <b>Single worker (24 – 30)</b>                  |
| <b>Couple with dependent children (31 – 50)</b> |
| <b>Couple with adult children (51 – 60)</b>     |
| <b>Financial independent couple (61 – 85)</b>   |

# Setting Goals

# **Why set goals?**

**To avoid being biased short-sighted**

**... to increase the probability of higher levels of happiness**

**Tracking activity against goals builds a sense of achievement**

# Types of goals

## 1. Life goals

**Not primarily related to career, dwelling or finances**

Relationships, living circumstances, travel, lifestyle, family, character ...

## 2. Career goals

**Related to your work and personal income (Unit 4)**

Type of organisation, type of work, position (seniority), income ...

## 3. Dwelling goals

**Related to your primary dwelling (Unit 5)**

Moving out of home, renting a property, buying first home, second home, final home ...

## 4. Financial goals

**Related to your savings, investments and risk management (many Units)**

Savings targets, debt levels, superannuation, insurance, tax, investments ....

# **Ideas that are helpful as you set goals**

- 1. Goals should be consistent with your values**
- 2. Spend regular 'quiet time' dreaming about the future**
- 3. We are biased in our beliefs about future happiness**
- 4. Identify long-term goals and write them down**
- 5. Identify interim short and medium-term goals as 'stepping stones'**
- 6. Meditate on your goals and actively pursue them**
- 7. Be flexible when things don't work out**
- 8. Regularly seek advice from a mentor or adviser (age > 40)**

# **Some ideas that are possibly unhelpful**

- 1. Just 'follow the rules' and things will work out**
- 2. "No one tells me what to do ... I forge my own path!"**
- 3. Clever people can control their destiny**
- 4. I can achieve my dreams if I just work harder**
- 5. "Don't worry! Be happy!"**
- 6. Being over-confident in your ability to succeed**
- 7. Being under-confident in your ability to succeed**
- 8. Dishonesty is a good way to 'get ahead'**

# Goals should be SMART

**S**pecific                      detailed enough so that you can visualise it

**M**easurable                      quantified so you can assess progress

**A**chievable                      realistic given your skills and resources

**R**elevant                      worthwhile pursuing

**T**ime-bound                      specify a date by which it will be achieved

# **1. Life goal examples**

**Get married to someone who is kind, caring and intelligent by 31 Dec 202X**

**Travel for 6 months through Europe after I graduate from my degree by 31 Jan 202X.**

**Meet with a UNSW counsellor at least 3 times before 31 Dec 202X to talk about my procrastination issues.**

## **2. Career goal examples**

**Progress my leadership skills by volunteering to lead on O-Week Activities in Feb 202X and to get feedback on my leadership from other team members**

**Complete the Speechcraft course at Toastmasters by 31 Dec 202X**

**Secure a 3 month contract after completing my degree in Sydney by 30 Jun 202X earning \$X0,000 p.a. for one of the following 3 companies: A, B or C.**

**Complete Masters of Business Administration at AGSM (UNSW Business School)\* on a part-time basis (while working) by 31 Dec 202X to get me ready for senior management and leadership positions at work**

\* A shameless cross-promotion for our MBA courses! I teach Finance on that program so I'll see you there.

# **3. Dwelling goal examples**

**Buy a 2 bedroom apartment in Liverpool for less than \$X00,000 by 31 Dec 202X**

**Buy a 4 bedroom home in SW Sydney for less than \$X,X00,000 by 31 Dec 203X**

**Downsize to a 2 bedroom apartment in Manly for less than \$X,X00,000 by 31 Dec 206X**

# **4. Financial goal examples**

**Create a minimum savings buffer of \$10,000 in my bank account by 31 Dec 202X.**

**Build financial slack of \$20,000 in my savings account by 31 Dec 202X for peace of mind.**

**Save \$X0,000 for a deposit on my first property by 31 Dec 202X.**

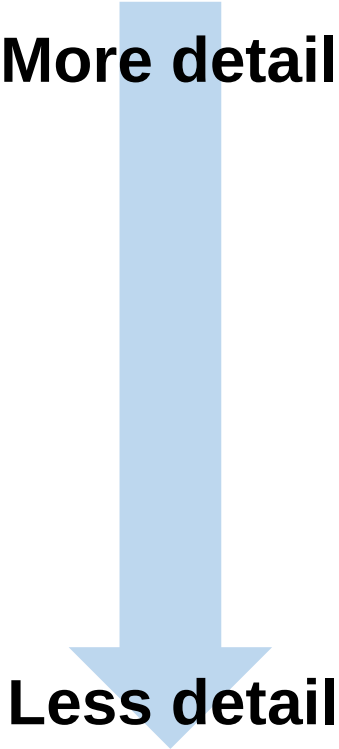
**Accumulate \$X0,000 in superannuation by age 2X.**

**Protect my personal income through appropriate life and health insurance**

**Protect my spouse and children from the financial consequences of my death or disability**

# Goals for different life stages

| LIFE PATH 1                              | LIFE PATH 2                               |
|------------------------------------------|-------------------------------------------|
| Student (19 – 23)                        | Student (19 – 23)                         |
| Single worker (24 – 30)                  | Single worker (24 – 30)                   |
| Couple with dependent children (30 – 50) | Start first business (31 – 50)            |
| Couple with adult children (50 – 60)     | Start second business (51 – 60)           |
| Financial independent couple (60 – 85)   | Financial independent investors (60 – 85) |



**Why does setting goals increase happiness?**

# **1. Provides a sense of purpose**

## **2. Allows us to manage short-term biases**

### **3. Pursuing goals develops self-discipline**

## **4. Planning process uncovers new alternatives**

## **5. Increases probability of achieving goals**

## **6. Anticipation = free happiness**

**7. Goals = happy parents = less intervention**

# Q: Identify some 'draft' goals

For each life stage, identify some life, career, dwelling and financial goals

| LIFE STAGE                                  | LIFE | CAREER | DWELLING | FINANCIAL |
|---------------------------------------------|------|--------|----------|-----------|
| Student<br>(19 – 23)                        |      |        |          |           |
| Single worker<br>(24 – 30)                  |      |        |          |           |
| Couple with dependent children<br>(30 – 50) |      |        |          |           |
| Couple with adult children<br>(50 – 60)     |      |        |          |           |
| Financial independent couple<br>(60 – 85)   |      |        |          |           |

# **Nominal and Real Rates of Return**

# Price inflation

**The price of goods and services usually increases over time**

... because growth in aggregate demand exceeds growth in aggregate supply

**This is called 'price inflation' and affects the cost of living**

**It is measured by the Consumer Price Index (CPI)**

CPI is measured by the Australian Bureau of Statistics

# Interest rates and inflation

Reserve Bank of Australia (RBA) influences interest rates

They set overnight 'cash rate' for borrowing and lending between banks

All interest rates are based on cash rate + risk premium

The main objective of the RBA is stable prices

The target rate of inflation is 2% to 3% per annum

They are very good at doing this!

In this course, we will assume long-term inflation = 2.5% per annum

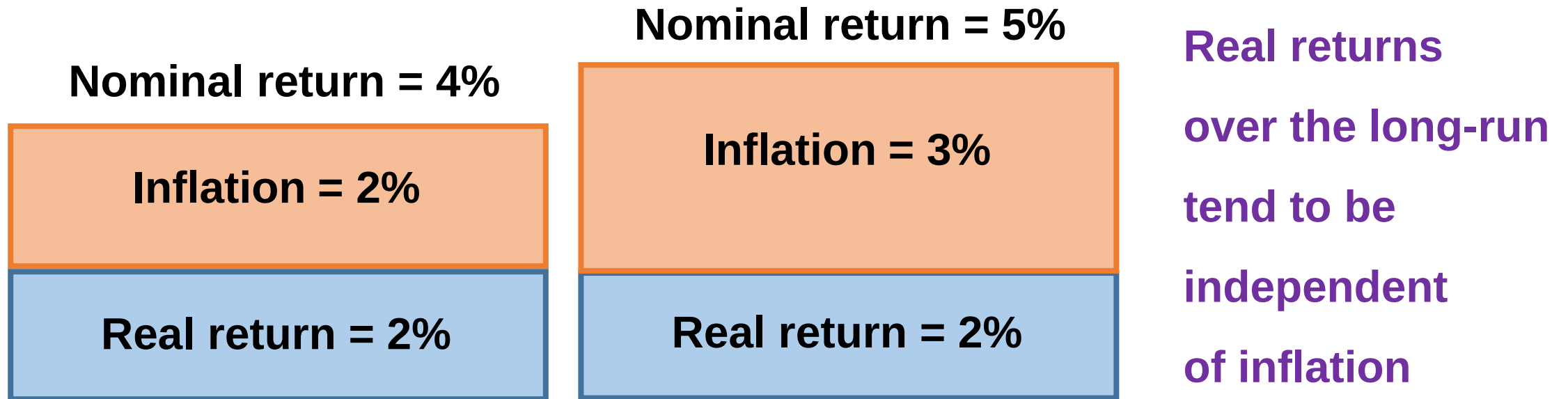
# Fisher Effect from Economics

Describes relationship between inflation, real and nominal rates of return

**Nominal rate of return\*  $\approx$  Real rate of return + inflation rate**

Nominal rate of return = Actual rate of return earned by an investment

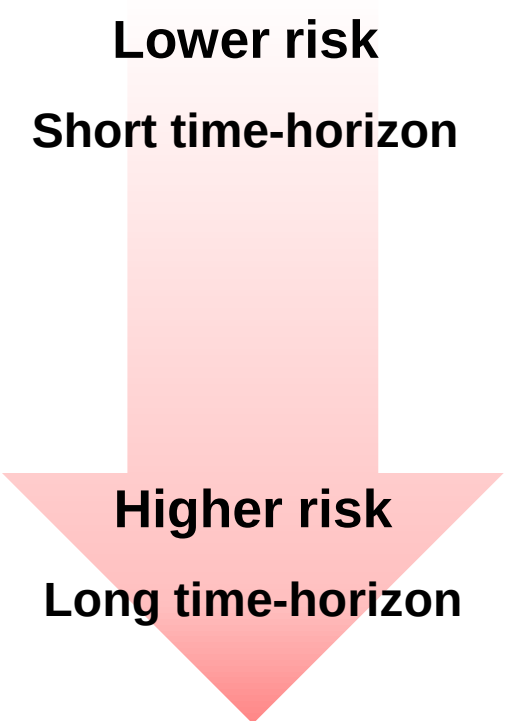
Real rate of return = Actual rate of return controlled for changes in the cost of living (inflation)



\* The correct formula is  $(1 + \text{nominal return}) = (1 + \text{real return}) \times (1 + \text{inflation rate})$  but there is little difference if inflation is low (2-3% p.a.)

\*\* This comes from 'Monetary Neutrality' which means that changes in the money supply (and inflation) affect nominal values and not real values.

# Asset categories

| Asset category          | Asset category                            |                                                                                                                                                                                                    |
|-------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Cash                 | Transaction, Savings Account or CMT       |  <p><b>Lower risk</b><br/><b>Short time-horizon</b></p> <p><b>Higher risk</b><br/><b>Long time-horizon</b></p> |
| 2. Fixed Interest       | Term deposits, debentures or bonds        |                                                                                                                                                                                                    |
| 3. Residential Property | Apartment, townhouse, duplex or house     |                                                                                                                                                                                                    |
| 4. Commercial Property  | Shop, office, Listed Property Trust (LPT) |                                                                                                                                                                                                    |
| 5. Australian Shares    | Australian Exchange Traded Fund (ETF)     |                                                                                                                                                                                                    |
| 6. International Shares | Global Exchange Traded Fund (ETF)         |                                                                                                                                                                                                    |

**Cryptocurrency (bitcoin), commodities (gold) and foreign exchange are speculative investments**

CMT is a Cash Management Trust. These are offered by investment banks such as Macquarie Bank and are popular with financial advisers  
A 'speculative investment' is a euphemism for 'gambling'. This is because they generate no cash flows. More on this in Units 9 and 10.

# Asset categories

| Asset category          | Asset category                            | Time horizon |
|-------------------------|-------------------------------------------|--------------|
| 1. Cash                 | Transaction, Savings Account or CMT       | 0 – 1 years  |
| 2. Fixed Interest       | Term deposits, debentures or bonds        | 1 – 5 years  |
| 3. Residential Property | Apartment, townhouse, duplex or house     | 5+ years     |
| 4. Commercial Property  | Shop, office, Listed Property Trust (LPT) | 5+ years     |
| 5. Australian Shares    | Australian Exchange Traded Fund (ETF)     | 5+ years     |
| 6. International Shares | Global Exchange Traded Fund (ETF)         | 5+ years     |

**Cryptocurrency (bitcoin), commodities (gold) and foreign exchange are speculative investments**

CMT is a Cash Management Trust. These are offered by investment banks such as Macquarie Bank and are popular with financial advisers  
A 'speculative investment' is a euphemism for 'gambling'. This is because they generate no cash flows. More on this in Units 9 and 10.

# Asset categories real and nominal returns

|    | Asset category       | Real return |   | Inflation |   | Nominal return |
|----|----------------------|-------------|---|-----------|---|----------------|
| 1. | Cash                 | 0.5%        | + | 2.5%      | = | 3.0%           |
| 2. | Fixed Interest       | 1.5%        | + | 2.5%      | = | 4.0%           |
| 3. | Residential Property | 5.0%        | + | 2.5%      | = | 7.5%           |
| 4. | Commercial Property  | 6.0%        | + | 2.5%      | = | 8.5%           |
| 5. | Australian Shares    | 6.0%        | + | 2.5%      | = | 8.5%           |
| 6. | International Shares | 6.0%        | + | 2.5%      | = | 8.5%           |

These are assumed long-term future expected rates of return for this course. The derivation will be discussed further in Units 9 and 10. They assume very low investment management fees and do not include the effects of income tax on investment returns.

# Decomposing nominal total returns

**Nominal returns = total returns = price changes + income**

## Price gains

- Change in bond prices (fixed interest)

- Change in property prices

- Change in share prices

- Change in unit prices

## Income

- Interest on savings or term deposits

- Coupons on bonds (fixed interest)

- Rent on properties

- Dividends on shares

- Distributions on managed funds or exchange traded funds (ETFs)

# Asset categories price gains and income

|    | Asset category       | Price gains |   | Income |   | Nominal return |
|----|----------------------|-------------|---|--------|---|----------------|
| 1. | Cash                 | 0.0%        | + | 3.0%   | = | 3.0%           |
| 2. | Fixed Interest       | 0.0%        | + | 4.0%   | = | 4.0%           |
| 3. | Residential Property | 5.0%        | + | 2.5%   | = | 7.5%           |
| 4. | Commercial Property  | 5.0%        | + | 3.5%   | = | 8.5%           |
| 5. | Australian Shares    | 5.0%        | + | 3.5%   | = | 8.5%           |
| 6. | International Shares | 6.5%        | + | 2.0%   | = | 8.5%           |

These are assumed long-term future expected rates of return for this course. The derivation will be discussed further in Units 9 and 10. They assume very low investment management fees and do not include the effects of income tax on investment returns.

# **Savings Calculations**

# **Some savings goals**

## **Short to medium term (0 to 5 years)**

**Financial buffer**

**Financial slack**

**Moving out of home**

**Motor vehicle**

**Overseas holiday**

**Deposit on property**

## **Long term (5+ years)**

**Debt reduction on property**

**Financial independence (super)**

**Children's school fees**

# Nominal or real rates of return

For short or medium-term financial goals (0 to 5 years) ...

... best to use nominal returns (including inflation)

For long term financial goals (5+ years) ...

... best to use real returns (excluding inflation)

In the following slides I will just use an arbitrary 5% for simplicity!

# Quote

The most powerful force in  
the universe is  
**compound interest**

*Attributed to Albert Einstein*

# **Compound interest = future value**

**Remember simple and compound interest from school?**

## **Simple interest**

Interest is calculated on principal amount saved only

## **Compound interest**

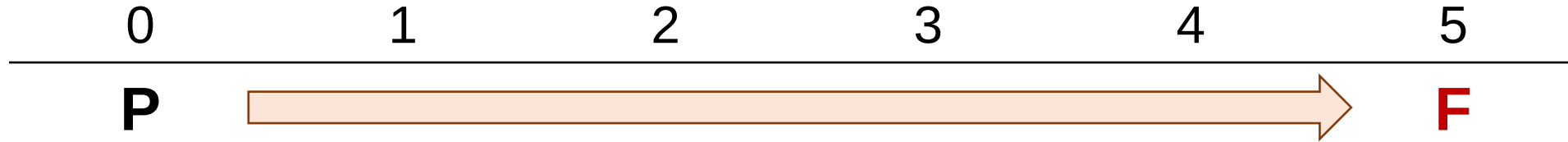
Interest is calculated based on both principal and past interest

**In this course we will always use compound interest**

*Let's have some fun with*

# **financial mathematics**

# Future value of single cash flow



$$F_n = P \times (1 + r)^n$$

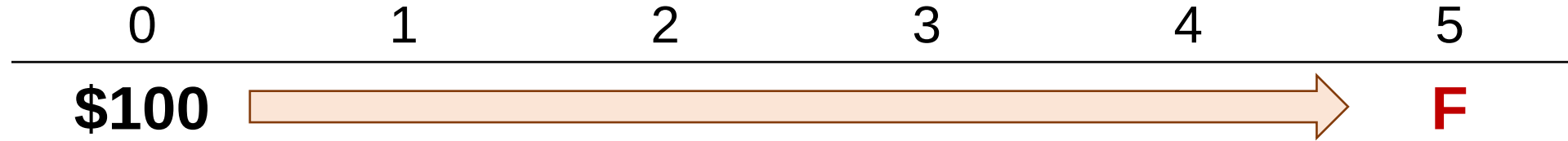
**‘P’ is the cash flow that is being moved to the right**

**‘n’ is the number of periods it is being moved**

**‘r’ is the effective rate of return per period** (time value of money)

**‘F’ is value of the cash flow ‘P’ after it has been moved ‘n’ periods right**

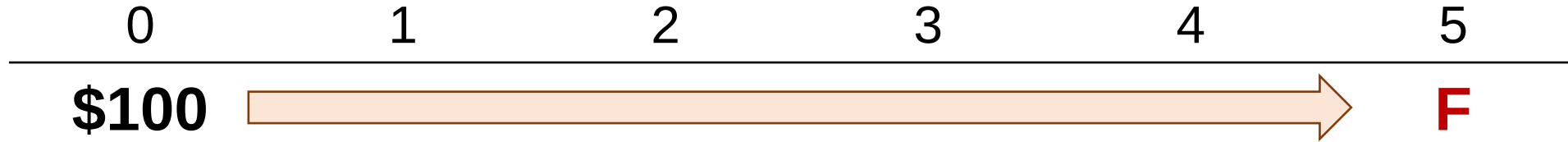
# Future value of single cash flow example



$$F_n = P \times (1 + r)^n$$

How much will we have in 5 years if we invest \$100 now at 5% p.a.?

# Future value of single cash flow example



$$F_n = P \times (1 + r)^n$$

How much will we have in 5 years if we invest \$100 now at 5% p.a.?

$$\begin{aligned} F_5 &= 100 \times (1 + 0.05)^5 \\ &= 100 \times 1.05^5 \\ &= \$127.63 \end{aligned}$$

Calculator

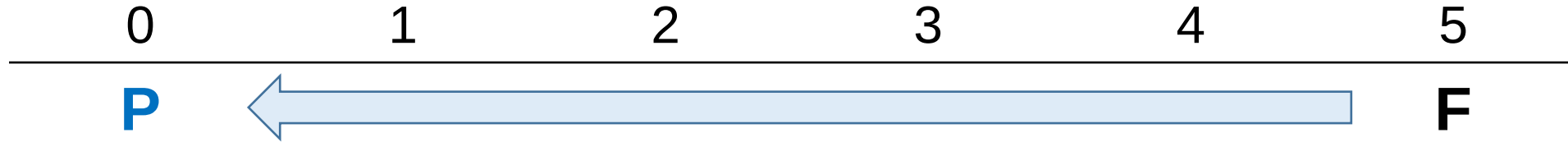
$$100 \times 1.05 \times^{\blacksquare} 5 \rightarrow =$$

Spreadsheet

$$= 100 * 1.05 ^ 5$$

Notice the value increases as we move it right

# Present value of single cash flow



$$P = \frac{F_n}{(1 + r)^n}$$

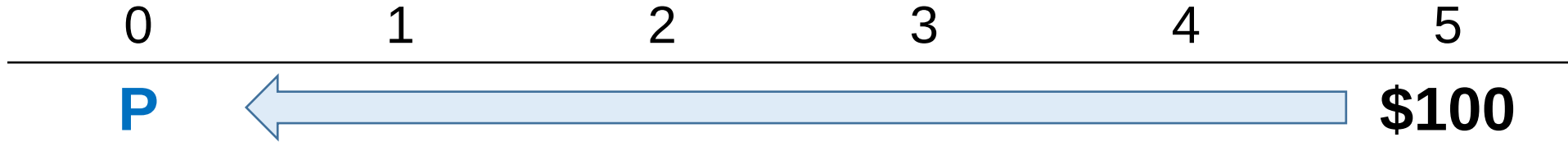
**'F'** is the cash flow that is being moved to the left

**'n'** is the number of periods it is being moved

**'r'** is the effective rate of return per period (time value of money)

**'P'** is value of the cash flow 'F' after it has been moved 'n' periods left

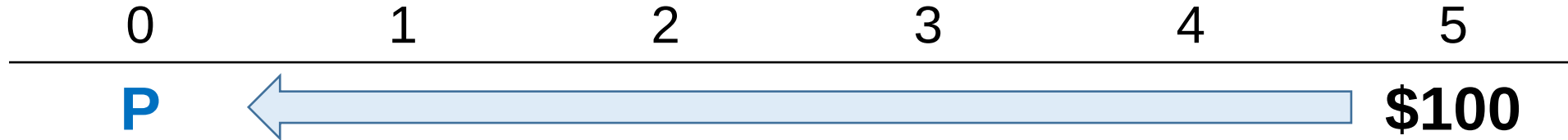
# Present value of single cash flow example



$$P = \frac{F_n}{(1 + r)^n}$$

How much do we invest now to have \$100 in 5 years invested at 5% p.a.?

# Present value of single cash flow example



$$P = \frac{F_n}{(1 + r)^n}$$

How much do we invest now to have \$100 in 5 years invested at 5% p.a.?

$$P = \frac{100}{1.05^5}$$

$$= \$78.35$$

Calculator

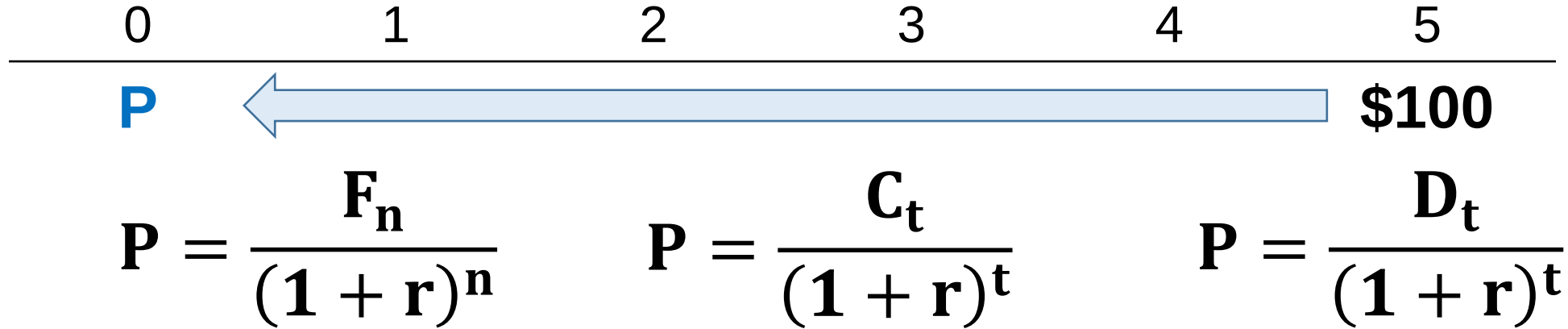
$$100 \div 1.05 \times^5 \rightarrow =$$

Spreadsheet

$$= 100 / 1.05 ^ 5$$

See how the value decreases as we move it left? In later Units we will call this 'discounting' cash flows

# In future Units we may use other letters



**‘C’ could mean a cash flow from an investment**

**‘D’ could mean a dividend on a share**

**‘t’ means that we are bumping the cash flow back ‘t’ periods**

**... either way, it’s the same formula!**

# What is an annuity?

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 |
|   | C | C | C | C | C |

**An annuity is a set of regular cash flows**

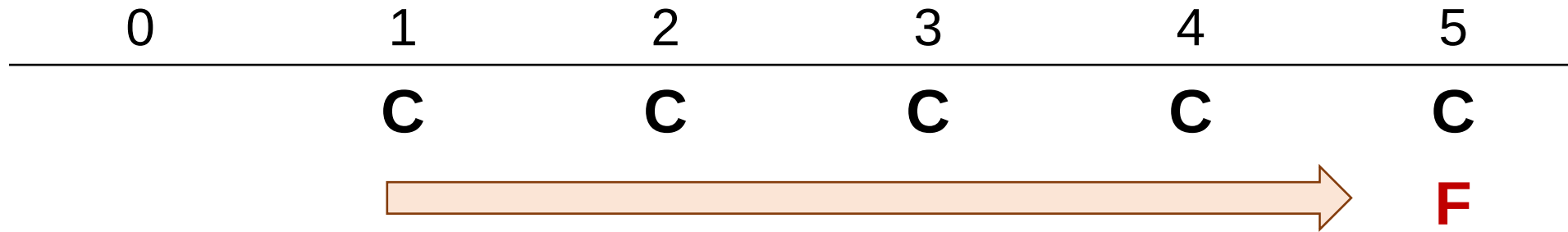
**For an 'ordinary' annuity**

**... the first cash flow occurs at the end of the first period**

**... and the cash flows are always level (not growing)**

**Useful for projects with level cash flows, loans and financial products**

# Future value of an annuity



We are calculating the value of the cash flows (with first cash flow at end of first period)

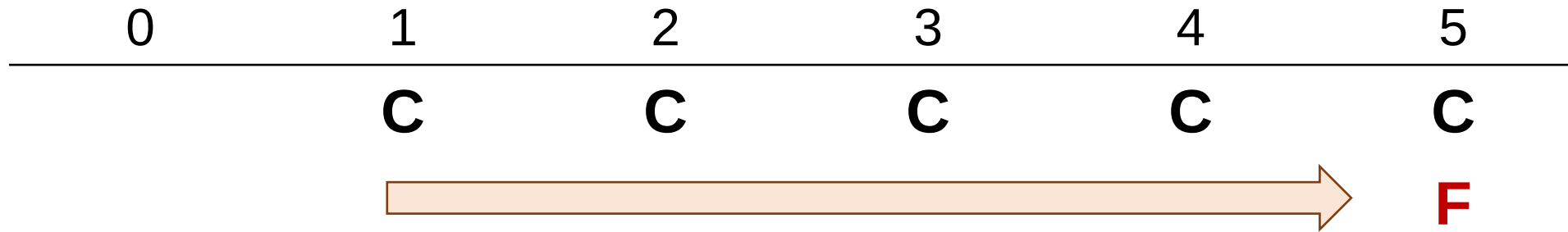
... at the **END** of the regular series of payments

Here the regular cash flow is 'C'

there are 5 cash flows and we are moving them to  $t = 5$

to calculate the future value of the annuity 'F'

# Future value of an annuity formula



$$F_n = C \times \left[ \frac{(1 + r)^n - 1}{r} \right]$$

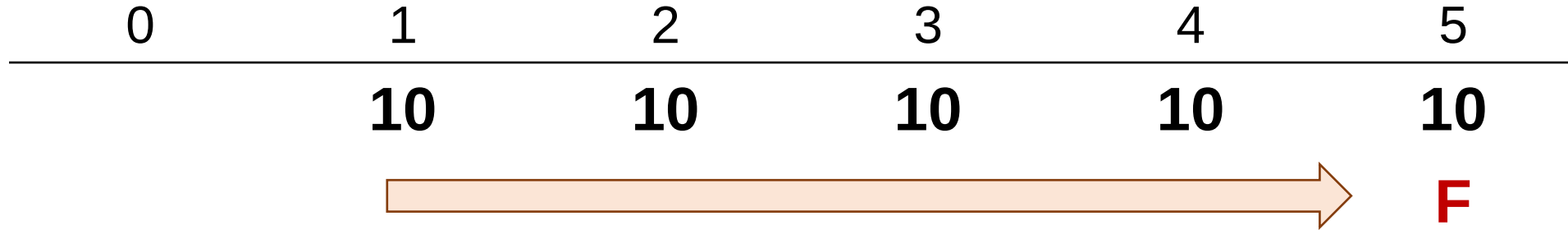
**‘C’** are the regular cash flows that are being moved to the right

**‘n’** is the number of cash flows (with first cash flow at end of first period)

**‘r’** is the effective rate of return per period (time value of money)

**‘F’** is value of the regular series of ‘n’ cash flows of ‘C’ at the end (right)

# Future value of an annuity example

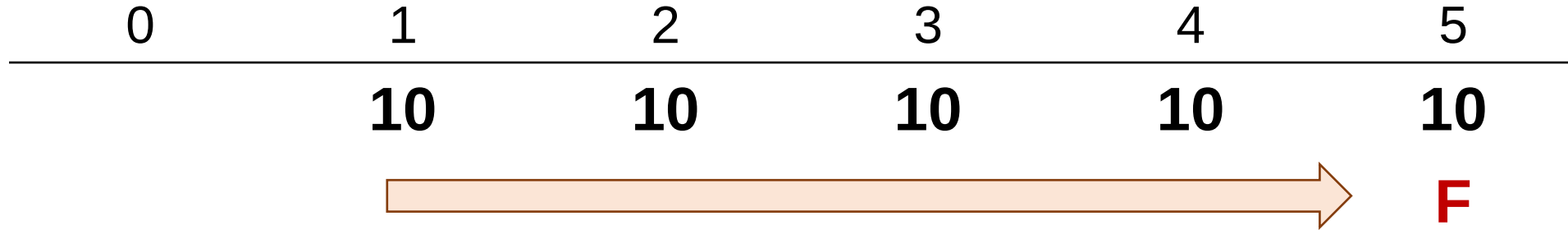


$$F_n = C \times \left[ \frac{(1 + r)^n - 1}{r} \right]$$

How much will we have in 5 years if we invest \$10 per year\* at 5% p.a.?

\* By default you should assume that the cash flows occur at the end of each period (unless otherwise specified)

# Future value of an annuity example



$$F_n = C \times \left[ \frac{(1 + r)^n - 1}{r} \right]$$

How much will we have in 5 years if we invest \$10 per year\* at 5% p.a.?

$$F_n = 10 \times \left[ \frac{1.05^5 - 1}{0.05} \right]$$

$$= \$55.26$$

Calculator

$$10 \times ( 1.05 \text{ x } ^5 \rightarrow -1 ) \div 0.05 =$$

Spreadsheet

$$= 10 * ( 1.05 ^ 5 - 1 ) / 0.05$$

\* By default you should assume that the cash flows occur at the end of each period (unless otherwise specified)

# We can re-arrange this formula to solve for C

$$\mathbf{F} = \mathbf{C} \times \left[ \frac{(\mathbf{1} + \mathbf{r})^{\mathbf{n}} - \mathbf{1}}{\mathbf{r}} \right]$$

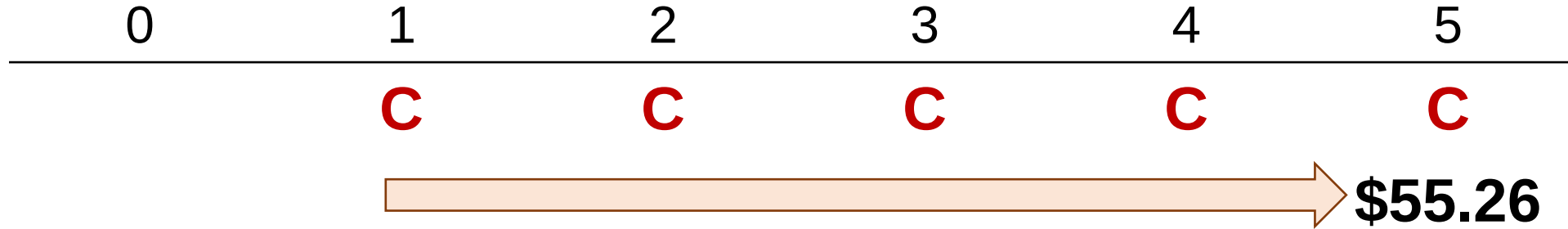
Now divide both sides by that big horrible thing in the square brackets

$$\mathbf{F} \div \left[ \frac{(\mathbf{1} + \mathbf{r})^{\mathbf{n}} - \mathbf{1}}{\mathbf{r}} \right] = \mathbf{C} \times \left[ \frac{(\mathbf{1} + \mathbf{r})^{\mathbf{n}} - \mathbf{1}}{\mathbf{r}} \right] \div \left[ \frac{(\mathbf{1} + \mathbf{r})^{\mathbf{n}} - \mathbf{1}}{\mathbf{r}} \right]$$

Anything divided by itself = 1 so ...

$$\mathbf{F} \div \left[ \frac{(\mathbf{1} + \mathbf{r})^{\mathbf{n}} - \mathbf{1}}{\mathbf{r}} \right] = \mathbf{C} \quad \begin{array}{c} \text{Then} \\ \text{swapping} \\ \text{sides} \end{array} \quad \mathbf{C} = \mathbf{F} \div \left[ \frac{(\mathbf{1} + \mathbf{r})^{\mathbf{n}} - \mathbf{1}}{\mathbf{r}} \right]$$

# Future value of an annuity payment example

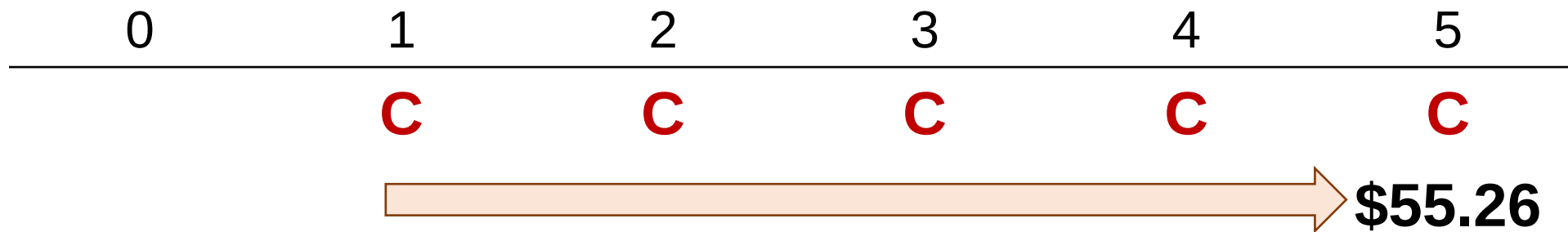


$$C = F_n \div \left[ \frac{(1 + r)^n - 1}{r} \right]$$

How much to invest each year\* to have \$55.26 in 5 years at 5% p.a.?

\* By default you should assume that the cash flows occur at the end of each period (unless otherwise specified)

# Future value of an annuity payment example



$$C = F_n \div \left[ \frac{(1 + r)^n - 1}{r} \right]$$

How much to invest each year\* to have \$55.26 in 5 years at 5% p.a.?

$$C = 55.26 \div \left[ \frac{1.05^5 - 1}{0.05} \right]$$

$$= \$10.00$$

Calculator

$$55.26 \div ( ( 1.05 \times^{\square} 5 \rightarrow -1 ) \div 0.05 ) =$$

Spreadsheet

$$= 55.26 / ( ( 1.05 ^ 5 - 1 ) / 0.05 )$$

\* By default you should assume that the cash flows occur at the end of each period (unless otherwise specified)

# **Q: Saving for a holiday**

**How much will you need to set aside now for a \$3,000 holiday in 2 years if the interest rate is 6% p.a. calculated and paid monthly (0.5% per month)?**

# Q: Saving for a holiday

How much will you need to set aside now for a \$3,000 holiday in 2 years if the interest rate is 6% p.a. calculated and paid monthly (0.5% per month)?

$$P = \frac{F_n}{(1 + r)^n}$$

$$= \frac{3,000}{1.005^{24}}$$

$$= \$2,661.56$$

Calculator

$$3000 \div 1.005 \times^{24} \rightarrow =$$

Spreadsheet

$$= 3000 / 1.005 ^{24}$$

# **Q: The Power of Time**

**A 20 year old starts saving \$500 every month and invests it at a return of 6% per year (0.5% per month).**

**How much will they have at age 60 (after 480 months)?**

# Q: The Power of Time

A 20 year old starts saving \$500 every month and invests it at a return of 6% per year (0.5% per month).

How much will they have at age 60 (after 480 months)?

$$\begin{aligned} F_n &= C \times \left[ \frac{(1 + r)^n - 1}{r} \right] \\ &= 500 \times \left[ \frac{1.005^{480} - 1}{0.005} \right] \\ &= \$995,745 \end{aligned}$$

Calculator

$$500 \times (1.005^{480} - 1) \div 0.005 =$$

Spreadsheet

$$= 500 * (1.005^{480} - 1) / 0.005$$

# **Q: Creating Financial Independence**

**A 20 year old has decided that they need \$800,000 to be financial independent at age 60 (40 years). They can invest at return of 6% per year. How much do they need to save each year to achieve this goal?**

# Q: Creating Financial Independence

A 20 year old has decided that they need \$800,000 to be financial independent at age 60 (40 years). They can invest at return of 6% per year. How much do they need to save each year to achieve this goal?

$$C = F_n \div \left[ \frac{(1 + r)^n - 1}{r} \right]$$

$$= 800,000 \div \left[ \frac{1.06^{40} - 1}{0.06} \right]$$

$$= \$5,169$$

Calculator

$$800000 \div ( ( 1.06 \times^{40} - 1 ) \div 0.06 ) =$$

Spreadsheet

$$= 800000 / ( ( 1.06 ^{40} - 1 ) / 0.06 )$$

# **Personal Finance Strategy**

# What is strategy?

To use a military metaphor

... strategy is how you plan to win the war

To use a sporting metaphor

... it is how you plan to win the game

Using the concept of 'fit' and 'alignment'

... it is achieving optimal 'fit' between you and your environment

Alan Lafley and Roger Martin (2013), *Playing to win: How strategy really works*

Michael Porter (1996), 'What is Strategy?', *Harvard Business Review*

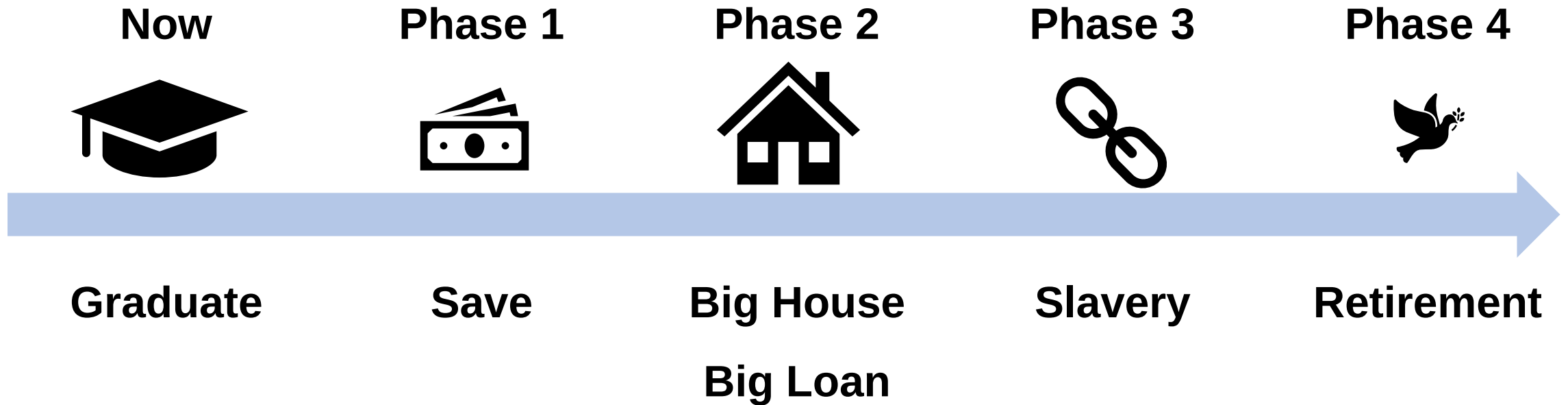
# **What is strategy for personal finance?**

**Achieving your long-term goals**

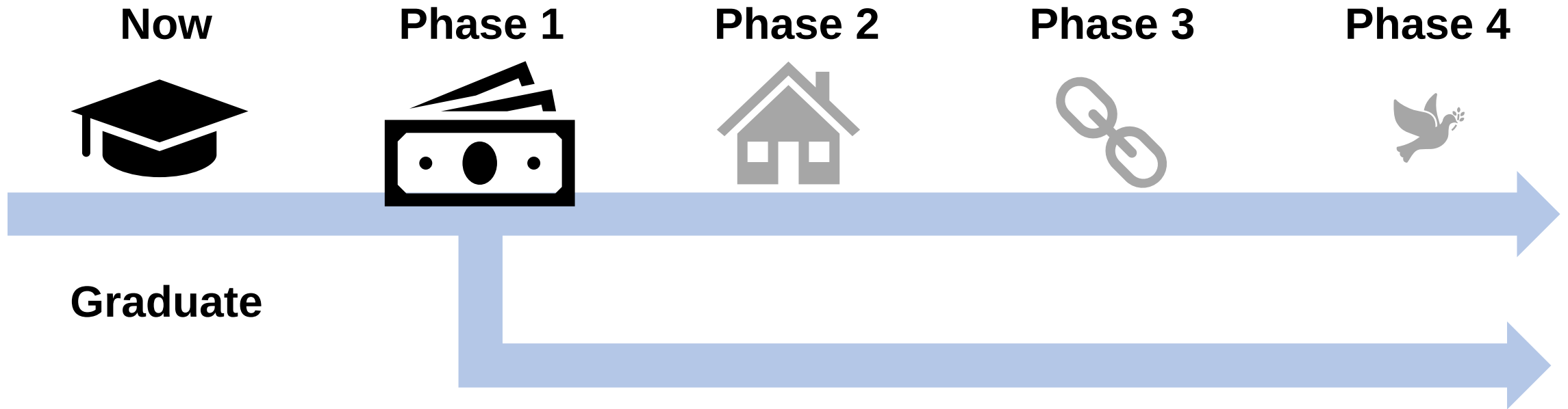
**in a way that brings life satisfaction to you and the people around you**

**... in a way that is consistent with your values**

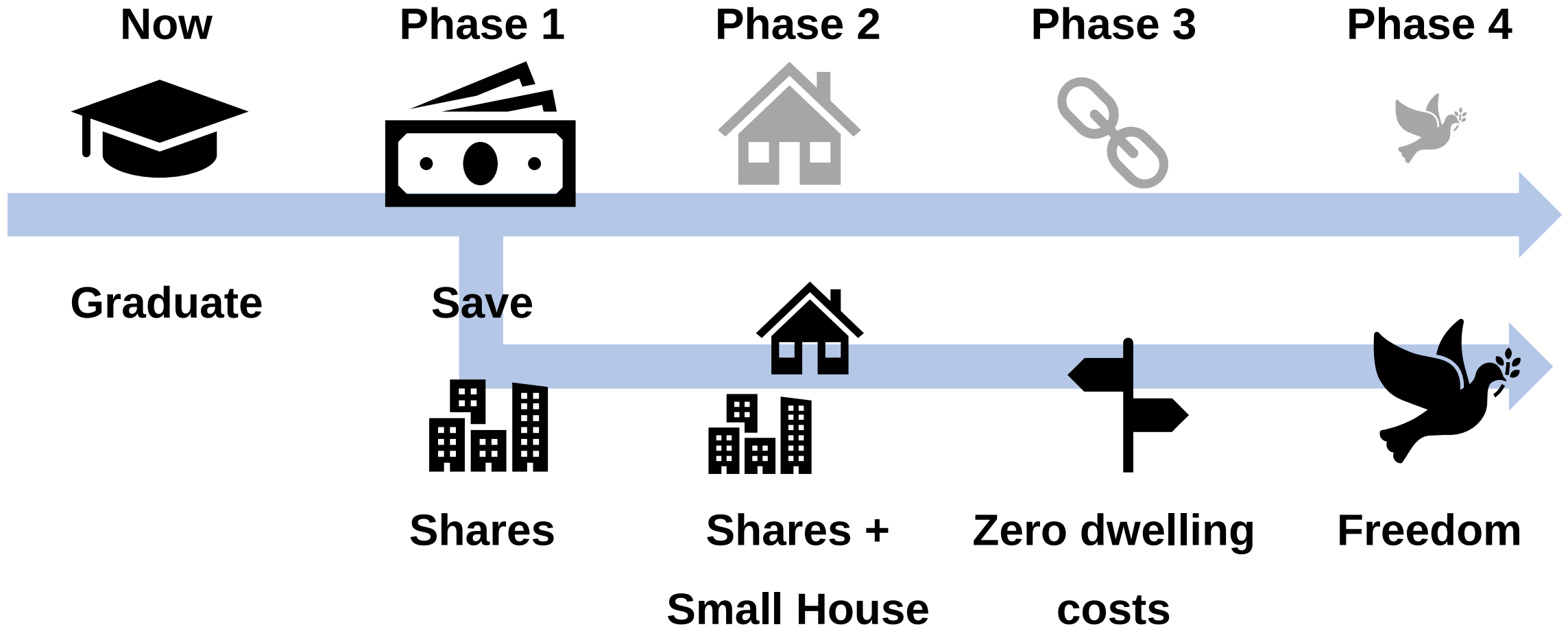
# This is what most people do ...



# An alternative strategy



# An alternative strategy example



# **Your Financial Plan**

**There are two sections that you need to write for your financial plan:**

## **2. Life Planning**

Life values and principles

Life stages

Goals for each life stage

## **3. Financial Strategy**

Your 'game plan' for achieving your long-term goals

We will keep revisiting and refining this strategy in future Units

... so it is just a rough 'first draft'

**More information under 'Financial Plan Instructions' document**

... on the course website under 'Financial Plan' section

# **Join us in a Drop-in Session**

**Each week we have a drop-in Q&A session on Zoom**

**Details are on the course website under ‘Getting Started Hub’**

**Bring along some questions that you have about the Unit.**

**Sessions are not recorded.**

**You are also welcome to ask questions on the ‘General Forums’**